

# **Graph Neural Networks for the Classification And Lead Identification of EGFR Inhibitors in Non-Small Cell Lung Cancer**



**A Thesis Submitted for the Partial Fulfillment of the  
Requirements for the Master of Science (M. Sc.)  
Degree in Data Science**

**Submitted to the  
Central Department of Mathematics /  
Kirtipur Campus  
Institute of Science And Technology  
Tribhuvan University  
NEPAL**

**By**

**Krishna Gaire  
Symbol No.: 7916015  
T.U. Registration No.: 5-3-28-49-2022**

**JULY, 2026**



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## **Declaration**

This thesis entitled “**Graph Neural Networks for the Classification And Lead Identification of EGFR Inhibitors in Non-Small Cell Lung Cancer**” which is being submitted to the Department of Mathematics, Kirtipur Campus, Institute of Science and Technology (IoST), Tribhuvan University, Nepal for the partial fulfillment of the requirement to the Master of Science Degree in Data Science has been carried out by me under the supervision of **[Name of Supervisor]** and co-supervision of **[Name of Co-Supervisor, if any]** at Central Department of Mathematics, Kirtipur Campus. This work is original and has not been submitted earlier in part or full in this or any other form to any other institution for the award of any degree.

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**Date: JULY,2026**

## **Supervisor's Recommendation**

This is to recommend that **Krishna Gaire** has carried out the thesis entitled “**Graph Neural Networks for the Classification And Lead Identification of EGFR Inhibitors in Non-Small Cell Lung Cancer**” for the partial fulfillment of the requirement of M. Sc. in Data Science under my/our supervision. To my/our knowledge, this work has not been submitted to any other institution for any academic degree. He has fulfilled all the requirements of the Department of Mathematics, Kirtipur Campus, Institute of Science and Technology (IoST), Tribhuvan University, Nepal for the submission of the thesis for the Master of Science degree in Data Science.

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**JULY, 2026**



**TRIBHUVAN UNIVERSITY**  
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**Letter of Approval**

The thesis work entitled “**Graph Neural Networks for the Classification And Lead Identification of EGFR Inhibitors in Non-Small Cell Lung Cancer**” submitted at the Institute of Science and Technology, Department of Mathematics, Tribhuvan University by **Krishna Gaire**, University Registration Number ....., has been accepted for the partial fulfillment of Master in Data Science.

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---

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## Abstract

Non-small cell lung cancer (NSCLC) accounts for approximately 85% of all lung cancer cases globally, with epidermal growth factor receptor (EGFR) mutations representing one of the most clinically significant oncogenic drivers. EGFR-targeted tyrosine kinase inhibitors have transformed the treatment landscape for EGFR-mutant NSCLC; however, acquired resistance and the high cost of therapy remain major clinical challenges, particularly in resource-limited South Asian settings. This study presents a graph neural network (GNN)-based computational framework for the classification of EGFR inhibitors and the prospective virtual screening of bioactive compounds from traditional Nepali medicinal plants.

A curated dataset of 10,523 unique compounds with experimentally validated EGFR bioactivity (74% active) was assembled from ChEMBL, standardised with RDKit, and represented as molecular graphs using 29-dimensional atom and 12-dimensional bond features. Seven classification models were benchmarked under a rigorous Bemis-Murcko scaffold split: five GNN architectures (GCN, GAT, GIN, AttentiveFP, and GraphSAGE) and two baselines (Random Forest and a multi-layer perceptron on 2048-bit ECFP4 fingerprints). The fingerprint baselines led on the scaffold test set (Random Forest AUROC 0.8827, MLP 0.8547), ahead of the graph neural networks (0.8087–0.8454, GCN strongest); Bayesian hyperparameter optimisation with Optuna raised AttentiveFP’s validation AUROC to 0.8632 but it did not surpass the baselines, indicating that well-tuned fingerprint models remain highly competitive on a scaffold-split dataset of this size.

The two best-performing models—the best graph network (GCN) and the best overall classifier (Random Forest)—were then applied to a virtual screening library of 535 natural product compounds curated from seven traditional Nepali medicinal plant species (*Swertia chirayita*, *Rhododendron arboreum*, *Nardostachys jatamansi*, *Terminalia chebula*, *Berberis aristata*, *Ocimum sanctum*, and *Tinospora cordifolia*) via the COCONUT 2.0 database. Compounds were ranked by predicted activity and assessed for drug-likeness using Lipinski’s Rule of Five and ADMET-proxy descriptors (QED, TPSA). The screening hit list proved strongly model-dependent—the Random Forest predicted 181 actives and the GCN only 24—exposing the earlier tuned-AttentiveFP screen as over-confident. Despite this, both best models converged on the same drug-like chemotype, the flavones: **apigenin** (*Ocimum sanctum*) was the single highest-

confidence Lipinski-compliant lead (Random Forest predicted probability 0.913, QED 0.632), while **luteolin**, predicted active by both models, was the top cross-model consensus candidate.

This work shows that model-consensus virtual screening can identify actionable, drug-like flavonoid lead candidates from Nepali ethnomedicinal compound libraries using an entirely open-source, CPU-only pipeline, while underscoring that natural-product screening predictions are strongly model-dependent and require cross-model consensus and experimental validation.

**Keywords:** *EGFR inhibitors, graph neural networks, virtual screening, Nepali medicinal plants, drug discovery.*

## शोधसार

फोक्सोको क्यान्सर विश्वव्यापी रूपमा सबैभन्दा बढी मृत्युदर भएको क्यान्सरहरूमध्ये एक हो, जसमध्ये नन-स्मल सेल लङ क्यान्सर (NSCLC) सम्पूर्ण फोक्सोको क्यान्सरको लगभग ८५% हिस्सा ओगट्दछ। EGFR उत्परिवर्तन यस रोगको प्रमुख आणविक चालकको रूपमा पहिचान गरिएको छ, तथापि अवरोधकहरूप्रति प्राप्त प्रतिरोधक क्षमता एक प्रमुख चुनौती बनेको छ।

यस अध्ययनमा EGFR अवरोधकहरूको वर्गीकरण र सम्भावित नेतृत्व यौगिकहरू पहिचान गर्न ग्राफ न्युरल नेटवर्क (GNN) आधारित गणनात्मक ढाँचा प्रस्तुत गरिएको छ। ChEMBL डेटाबेसबाट प्रयोगात्मक रूपमा प्रमाणित EGFR जैव-क्रियाशीलता भएका १०,५२३ यौगिकहरू संकलन गरी, RDKit प्रयोग गरी स्तरीकृत गरिए र आणविक ग्राफको रूपमा प्रस्तुत गरियो। Bemis-Murcko स्क्रियाफोल्ड विभाजन अन्तर्गत पाँच GNN वास्तुकला (GCN, GAT, GIN, AttentiveFP, GraphSAGE) र दुई आधारभूत मोडेल (Random Forest, MLP) गरी सात मोडेलको तुलना गरियो। फिंगरप्रिन्ट-आधारित आधारभूत मोडेलहरू (Random Forest AUROC 0.8827, MLP 0.8547) ले सर्वोच्च प्रदर्शन देखाए, जबकि GNN हरू (0.8087–0.8454, जसमध्ये GCN सबैभन्दा बलियो) तुलनायोग्य रहे। Optuna मार्फत अनुकूलन गरिएको AttentiveFP ले 0.8632 को मान्यकरण AUROC प्राप्त गर्‍यो, तर यसले आधारभूत मोडेलहरूलाई उछिन्न सकेन।

अध्ययनका दुई उत्कृष्ट मोडेल (उत्कृष्ट ग्राफ नेटवर्क GCN र उत्कृष्ट वर्गीकरणकर्ता Random Forest) लाई COCONUT 2.0 डेटाबेसमार्फत सात परम्परागत नेपाली औषधीय वनस्पति प्रजातिहरू (*Sweretia chirayita*, *Rhododendron arboreum*, *Nardostachys jatamansi*, *Terminalia chebula*, *Berberis aristata*, *Ocimum sanctum*, *Tinospora cordifolia*) बाट संकलित ५३५ प्राकृतिक यौगिकहरूको भर्चुअल स्क्रिनिङमा प्रयोग गरियो। यौगिकहरूलाई पूर्वानुमानित सक्रियताका आधारमा क्रमबद्ध गरी Lipinski नियम र ADMET गुणहरू (QED, TPSA) अनुसार मूल्याङ्कन गरियो। स्क्रिनिङको नतिजा मोडेलमा निर्भर रह्यो--- Random Forest ले १८१ र GCN ले २४ सम्भावित-सक्रिय यौगिक पहिचान गरे। तथापि दुवै उत्कृष्ट मोडेलले फ्लामोन वर्गका यौगिकहरूमा सहमति जनाए। *Ocimum sanctum* बाट प्राप्त apigenin सबैभन्दा उच्च-विश्वासको Lipinski -अनुपालित यौगिक (Random Forest सम्भावना ०.९१३, QED ०.६३२) रह्यो, भने दुवै मोडेलले सक्रिय ठहर्याएको luteolin शीर्ष सहमति-यौगिक रह्यो। यस अध्ययनले GNN आधारित गहन अधिगम विधिले NSCLC उपचारका लागि नेपाली परम्परागत औषधीय वनस्पतिहरूबाट नयाँ नेतृत्व यौगिकहरू पहिचान गर्न प्रभावकारी भूमिका खेल्न सक्ने देखाएको छ।

**मुख्य शब्दहरू:** *EGFR, graph neural networks, AttentiveFP, virtual screening, Nepali medicinal plants, natural products, Bemis-Murcko scaffold split, non-small cell lung cancer, drug discovery*

## List of Abbreviations and Acronyms

| Abbreviation      | Full Form                                                           |
|-------------------|---------------------------------------------------------------------|
| ADMET             | Absorption, Distribution, Metabolism, Excretion, and Toxicity       |
| ATP               | Adenosine Triphosphate                                              |
| AUPRC             | Area Under the Precision–Recall Curve                               |
| AUROC             | Area Under the Receiver Operating Characteristic Curve              |
| C797S             | Cysteine-797 to Serine substitution (EGFR resistance mutation)      |
| ChEMBL            | Chemical European Molecular Biology Laboratory                      |
| COCONUT           | Collection of Open Natural prodUcTs (natural product database)      |
| CPU               | Central Processing Unit                                             |
| ECFP              | Extended Connectivity Fingerprint                                   |
| ECFP4             | Extended Connectivity Fingerprint, radius 2 (4-bond diameter)       |
| EGF               | Epidermal Growth Factor                                             |
| EGFR              | Epidermal Growth Factor Receptor                                    |
| ErbB              | Erythroblastic Leukaemia Viral Oncogene Homologue (receptor family) |
| Ex19del           | Exon 19 Deletion (EGFR sensitising mutation)                        |
| F1                | F1 Score (harmonic mean of precision and recall)                    |
| GAT               | Graph Attention Network                                             |
| GCN               | Graph Convolutional Network                                         |
| GIN               | Graph Isomorphism Network                                           |
| GNN               | Graph Neural Network                                                |
| GPU               | Graphics Processing Unit                                            |
| GraphSAGE         | Graph SAmple and aggreGatE                                          |
| GRU               | Gated Recurrent Unit                                                |
| HBA               | Hydrogen Bond Acceptors                                             |
| HBD               | Hydrogen Bond Donors                                                |
| HER2              | Human Epidermal Growth Factor Receptor 2 (ErbB2)                    |
| IC <sub>50</sub>  | Half-Maximal Inhibitory Concentration                               |
| IoST              | Institute of Science and Technology                                 |
| L858R             | Leucine-858 to Arginine substitution (EGFR mutation)                |
| MCC               | Matthews Correlation Coefficient                                    |
| ML                | Machine Learning                                                    |
| MLP               | Multi-Layer Perceptron                                              |
| MW                | Molecular Weight                                                    |
| NSCLC             | Non-Small Cell Lung Cancer                                          |
| pIC <sub>50</sub> | Negative Logarithm of IC <sub>50</sub>                              |
| QED               | Quantitative Estimate of Drug-likeness                              |

|        |                                                                      |
|--------|----------------------------------------------------------------------|
| ReLU   | Rectified Linear Unit                                                |
| RF     | Random Forest                                                        |
| Ro5    | Lipinski's Rule of Five                                              |
| SMILES | Simplified Molecular Input Line Entry System                         |
| T790M  | Threonine-790 to Methionine substitution (EGFR gate-keeper mutation) |
| TKI    | Tyrosine Kinase Inhibitor                                            |
| TPSA   | Topological Polar Surface Area                                       |
| TU     | Tribhuvan University                                                 |
| VS     | Virtual Screening                                                    |

## List of Symbols

| Symbol          | Description                                                   |
|-----------------|---------------------------------------------------------------|
| $\hat{p}$       | Predicted probability of activity from the classifier         |
| $d_n$           | Dimensionality of atom node feature vectors (29 in this work) |
| $d_e$           | Dimensionality of bond edge feature vectors (12 in this work) |
| $n$             | Number of compounds                                           |
| $\text{Log } P$ | Logarithm of the octanol–water partition coefficient          |
| $\text{\AA}$    | Ångström ( $10^{-10}$ m)                                      |
| $\text{\AA}^2$  | Square Ångström (area unit for TPSA)                          |
| Da              | Dalton (atomic mass unit)                                     |
| nM              | Nanomolar ( $10^{-9}$ mol/L)                                  |
| $\mu\text{M}$   | Micromolar ( $10^{-6}$ mol/L)                                 |

# List of Tables

|                  |                                                                                                                                                                                                                                                                                           |    |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Table 1:</b>  | Comparison of this thesis against representative published GNN-based EGFR inhibitor classification studies across three methodological dimensions. ✓ = fully addressed; ◦ = partially addressed; × = not addressed. . . . .                                                               | 26 |
| <b>Table 2:</b>  | Dataset partitioning using the Bemis-Murcko scaffold split. . . .                                                                                                                                                                                                                         | 30 |
| <b>Table 3:</b>  | Hyperparameter search space for AttentiveFP Optuna optimisation.                                                                                                                                                                                                                          | 33 |
| <b>Table 4:</b>  | Nepali medicinal plant species used for virtual screening. . . . .                                                                                                                                                                                                                        | 33 |
| <b>Table 5:</b>  | Dataset composition and class distribution across scaffold-split partitions. . . . .                                                                                                                                                                                                      | 35 |
| <b>Table 6:</b>  | Summary statistics of physicochemical properties of the curated dataset. . . . .                                                                                                                                                                                                          | 36 |
| <b>Table 7:</b>  | Performance comparison of all models on the scaffold-held-out test set (threshold = 0.50 for F1, MCC, Accuracy). . . . .                                                                                                                                                                  | 39 |
| <b>Table 8:</b>  | Optimal AttentiveFP hyperparameters identified by Optuna Bayesian optimisation over 20 trials, with the resulting performance. . . . .                                                                                                                                                    | 41 |
| <b>Table 9:</b>  | Composition of the Nepali medicinal plant screening library from COCONUT 2.0. . . . .                                                                                                                                                                                                     | 42 |
| <b>Table 10:</b> | Nepali plant virtual screening funnel under the three models, showing the number of compounds retained at each activity and drug-likeness stage. The three models disagree sharply, revealing the model-sensitivity of the screen. . . . .                                                | 43 |
| <b>Table 11:</b> | Top drug-like (Lipinski-compliant) Nepali plant leads nominated by the best overall classifier (Random Forest), ranked by predicted activity. Listed compounds are those bearing common (trivial) names. . . . .                                                                          | 45 |
| <b>Table 12:</b> | Top drug-like (Lipinski-compliant) Nepali plant leads nominated by the best graph neural network (GCN), ranked by predicted activity. The GCN is markedly more conservative: no compliant compound exceeds $\hat{p} = 0.66$ , and none reaches the $\hat{p} \geq 0.80$ threshold. . . . . | 45 |
| <b>Table 13:</b> | Predicted activity of the flavone scaffolds under the two best models. Only luteolin is predicted active ( $\hat{p} \geq 0.50$ ) by <i>both</i> models. . . . .                                                                                                                           | 45 |
| <b>Table 14:</b> | ChEMBL query parameters used to retrieve EGFR bioactivity data. . . . .                                                                                                                                                                                                                   | 57 |

|                  |                                                                                                                                                 |    |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Table 15:</b> | Step-by-step dataset curation statistics for the EGFR bioactivity dataset. . . . .                                                              | 57 |
| <b>Table 16:</b> | Composition of the Nepali medicinal plant screening library retrieved from COCONUT 2.0, by species. . . . .                                     | 59 |
| <b>Table 17:</b> | Canonical SMILES and physicochemical properties of the highest-confidence Lipinski-compliant lead (Random Forest). . .                          | 60 |
| <b>Table 18:</b> | Summary of virtual screening results from the two best models applied to the 535-compound Nepali medicinal plant library (COCONUT 2.0). . . . . | 60 |
| <b>Table 19:</b> | Final hyperparameters of the tuned AttentiveFP model selected by Optuna Bayesian optimisation. . . . .                                          | 61 |
| <b>Table 20:</b> | Performance comparison of GNN architectures and traditional ML baselines on the EGFR scaffold test set (Bemis-Murcko split). .                  | 61 |
| <b>Table 21:</b> | Node (atom) and edge (bond) feature vectors used as input to all GNN architectures. . . . .                                                     | 62 |
| <b>Table 22:</b> | Software packages, libraries, and databases used in this study. . .                                                                             | 63 |
| <b>Table 23:</b> | Computational hardware and software environment used for all experiments. . . . .                                                               | 63 |

# List of Figures

|                  |                                                                                                                                                                                                                                                                                                                                                                                |    |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Figure 1:</b> | Representation of a molecule as an attributed graph for graph neural network learning. The 2D structure of the EGFR inhibitor gefitinib (a) is converted into a molecular graph (b) in which each atom becomes a node and each bond an edge. Aromatic atoms are shown in green and non-aromatic atoms in blue; each atom and bond is encoded by a fixed-length feature vector. | 4  |
| <b>Figure 2:</b> | One step of graph neural network message passing. The central atom $v$ updates its representation by aggregating learned messages from its bonded neighbours $u_1, u_2, u_3$ . Repeating this over several layers lets each atom’s embedding incorporate progressively larger chemical neighbourhoods.                                                                         | 19 |
| <b>Figure 3:</b> | Five-phase computational pipeline for GNN-based EGFR inhibitor classification and Nepali medicinal plant virtual screening.                                                                                                                                                                                                                                                    | 28 |
| <b>Figure 4:</b> | Class balance of the curated EGFR dataset: 74.0% active and 26.0% inactive compounds across 10,523 unique molecules.                                                                                                                                                                                                                                                           | 36 |
| <b>Figure 5:</b> | Distribution of EGFR bioactivity ( $\text{pIC}_{50}$ ) for the curated dataset, coloured by activity label. The active and inactive subpopulations form two modes separated by the $\text{pIC}_{50} = 6.0$ decision threshold (dashed line, $\text{IC}_{50} = 1 \mu\text{M}$ ).                                                                                                | 37 |
| <b>Figure 6:</b> | Distributions of molecular weight (left) and calculated LogP (right) for a random sample of the curated dataset. Dashed lines mark the Lipinski thresholds ( $\text{MW} = 500 \text{ Da}$ , $\text{LogP} = 5$ ); most compounds cluster near these thresholds, characteristic of kinase inhibitors.                                                                            | 37 |
| <b>Figure 7:</b> | Bemis-Murcko scaffold diversity of the curated dataset. Left: number of compounds sharing each of the most frequent scaffolds. Right: the split between singleton scaffolds (a single compound) and multi-compound scaffolds, illustrating the long-tailed scaffold distribution that necessitates scaffold-based splitting.                                                   | 38 |
| <b>Figure 8:</b> | Performance of all seven models on the scaffold-held-out test set across AUROC, AUPRC, F1-score, and MCC. The Random Forest and MLP fingerprint baselines lead on AUROC, while the graph neural networks form a competitive cluster led by GCN.                                                                                                                                | 39 |
| <b>Figure 9:</b> | Composition of the Nepali medicinal plant screening library by species, totalling 535 unique natural product compounds retrieved from COCONUT 2.0.                                                                                                                                                                                                                             | 43 |

- Figure 10:** Virtual screening funnel for the Nepali medicinal plant library under the three models (tuned AttentiveFP, best GNN = GCN, best classifier = Random Forest), showing the number of compounds retained at each prioritisation stage. The sharp disagreement between models—AttentiveFP retaining 71 compounds at  $\hat{p} \geq 0.90$  versus 0 for GCN and 1 for Random Forest—illustrates the model-sensitivity of the screen. . . . . 44
- Figure 11:** Random Forest drug-like Nepali plant lead candidates in the predicted-activity versus drug-likeness (QED) plane. All shown candidates are Lipinski-compliant and marker size is proportional to molecular weight. Dashed lines mark the  $H_2$  thresholds ( $\hat{p} \geq 0.80$ , QED  $\geq 0.40$ ); apigenin and kaempferol clear both. . . . . 46

# Contents

|                                                                                                   |             |
|---------------------------------------------------------------------------------------------------|-------------|
| <b>Declaration</b>                                                                                | <b>ii</b>   |
| <b>Supervisor's Recommendation</b>                                                                | <b>iii</b>  |
| <b>Letter of Approval</b>                                                                         | <b>iv</b>   |
| <b>Acknowledgements</b>                                                                           | <b>v</b>    |
| <b>Abstract</b>                                                                                   | <b>vi</b>   |
| <b>शोधसार</b>                                                                                     | <b>viii</b> |
| <b>List of Abbreviations and Acronyms</b>                                                         | <b>ix</b>   |
| <b>List of Symbols</b>                                                                            | <b>xi</b>   |
| <b>List of Tables</b>                                                                             | <b>xiii</b> |
| <b>List of Figures</b>                                                                            | <b>xv</b>   |
| <b>1 Introduction</b>                                                                             | <b>1</b>    |
| 1.1 Background . . . . .                                                                          | 1           |
| 1.1.1 The Epidermal Growth Factor Receptor in NSCLC . . . . .                                     | 1           |
| 1.1.2 Generations of EGFR Tyrosine Kinase Inhibitors . . . . .                                    | 2           |
| 1.1.3 Computational Drug Discovery and Graph Neural Networks . . . . .                            | 4           |
| 1.2 Problem Statement . . . . .                                                                   | 5           |
| 1.2.1 Clinical and Translational Need . . . . .                                                   | 5           |
| 1.2.2 Computational Problem: Methodological Gaps in GNN-Based EGFR Inhibitor Prediction . . . . . | 5           |
| 1.3 Objectives . . . . .                                                                          | 6           |
| 1.3.1 General Objective . . . . .                                                                 | 6           |
| 1.3.2 Specific Objectives . . . . .                                                               | 6           |
| 1.4 Rationale . . . . .                                                                           | 6           |
| 1.5 Research Hypotheses . . . . .                                                                 | 7           |
| 1.6 Significance of the Study . . . . .                                                           | 7           |
| 1.7 Limitations . . . . .                                                                         | 8           |
| 1.8 Organisation of the Thesis . . . . .                                                          | 10          |

|          |                                                                 |           |
|----------|-----------------------------------------------------------------|-----------|
| <b>2</b> | <b>Literature Review</b>                                        | <b>11</b> |
| 2.1      | EGFR Kinase Biology and NSCLC                                   | 11        |
| 2.1.1    | Molecular Architecture of EGFR                                  | 11        |
| 2.1.2    | Oncogenic EGFR Mutations in NSCLC                               | 12        |
| 2.2      | Three Generations of EGFR TKIs and Resistance Mechanisms        | 13        |
| 2.2.1    | First-Generation Reversible TKIs                                | 13        |
| 2.2.2    | Second-Generation Irreversible TKIs                             | 13        |
| 2.2.3    | Third-Generation Covalent TKIs and the C797S Resistance Problem | 14        |
| 2.3      | Bioactivity Databases and the ChEMBL Database                   | 15        |
| 2.3.1    | Overview of the ChEMBL Database                                 | 15        |
| 2.3.2    | Bioactivity Endpoints and pIC <sub>50</sub> Conversion          | 15        |
| 2.3.3    | Data Quality Challenges                                         | 16        |
| 2.4      | Molecular Representation for Machine Learning                   | 16        |
| 2.4.1    | Hand-Crafted Fingerprint Descriptors                            | 16        |
| 2.4.1.1  | Extended Connectivity Fingerprints (ECFP)                       | 16        |
| 2.4.1.2  | Other Descriptor Types                                          | 17        |
| 2.4.2    | Graph-Based Molecular Representations                           | 17        |
| 2.4.3    | Graph Neural Networks: The Message-Passing Framework            | 18        |
| 2.5      | Graph Neural Network Architectures for Drug Discovery           | 18        |
| 2.5.1    | Graph Convolutional Network (GCN)                               | 18        |
| 2.5.2    | Graph Attention Network (GAT)                                   | 19        |
| 2.5.3    | Graph Isomorphism Network (GIN)                                 | 20        |
| 2.5.4    | AttentiveFP                                                     | 20        |
| 2.5.4.1  | Atom-Level Attention                                            | 20        |
| 2.5.4.2  | Molecule-Level Iterative Readout                                | 21        |
| 2.5.5    | GraphSAGE                                                       | 21        |
| 2.6      | Nepali Medicinal Plants and Natural Product Databases           | 22        |
| 2.6.1    | Traditional Medicinal Plants in Nepal                           | 22        |
| 2.6.2    | The COCONUT 2.0 Database                                        | 23        |
| 2.7      | Virtual Screening in Drug Discovery                             | 23        |
| 2.7.1    | Overview of Virtual Screening                                   | 23        |
| 2.7.2    | GNN-Based Virtual Screening                                     | 24        |
| 2.7.3    | Natural Product Libraries and Nepali Medicinal Plants           | 24        |
| 2.7.4    | ADMET Properties and Lead Prioritisation                        | 24        |
| 2.8      | Research Gap                                                    | 25        |
| <b>3</b> | <b>Materials and Methods</b>                                    | <b>27</b> |
| 3.1      | System Flow Diagram of the Computational Framework              | 27        |
| 3.1.1    | Overview of the Computational Framework                         | 27        |
| 3.2      | Data Acquisition                                                | 27        |
| 3.2.1    | Target Selection                                                | 27        |
| 3.2.2    | Bioactivity Data Retrieval                                      | 27        |
| 3.2.3    | Data Cleaning and Labelling                                     | 27        |
| 3.2.4    | Molecular Standardisation and Filtering                         | 29        |

|          |                                                                       |           |
|----------|-----------------------------------------------------------------------|-----------|
| 3.2.5    | Exploratory Data Analysis                                             | 29        |
| 3.3      | Molecular Graph Construction                                          | 29        |
| 3.3.1    | Graph Representation                                                  | 29        |
| 3.3.2    | Node Features                                                         | 29        |
| 3.3.3    | Edge Features                                                         | 30        |
| 3.4      | Data Splitting                                                        | 30        |
| 3.5      | Machine Learning Model Development                                    | 30        |
| 3.5.1    | Traditional Baseline Models                                           | 30        |
| 3.5.2    | Graph Neural Network Architectures                                    | 31        |
| 3.5.2.1  | Graph Convolutional Network (GCN)                                     | 31        |
| 3.5.2.2  | Graph Attention Network (GAT)                                         | 31        |
| 3.5.2.3  | Graph Isomorphism Network (GIN)                                       | 31        |
| 3.5.2.4  | GraphSAGE                                                             | 31        |
| 3.5.2.5  | AttentiveFP                                                           | 32        |
| 3.5.3    | Training Protocol                                                     | 32        |
| 3.5.4    | Model Evaluation Metrics                                              | 32        |
| 3.5.5    | Hyperparameter Tuning                                                 | 33        |
| 3.6      | Nepali Medicinal Plant Virtual Screening                              | 33        |
| 3.6.1    | Plant Species and Compound Library Curation                           | 33        |
| 3.6.2    | Inference and Prioritisation                                          | 34        |
| <b>4</b> | <b>Results and Discussion</b>                                         | <b>35</b> |
| 4.1      | Data Distribution Analysis                                            | 35        |
| 4.1.1    | Dataset Composition and Class Balance                                 | 35        |
| 4.1.2    | pIC <sub>50</sub> Distribution                                        | 35        |
| 4.1.3    | Physicochemical Property Analysis                                     | 36        |
| 4.1.4    | Scaffold Diversity Analysis                                           | 38        |
| 4.2      | Machine Learning Model Performance                                    | 38        |
| 4.2.1    | Individual Model Performance on the Scaffold Test Set                 | 38        |
| 4.2.1.1  | Random Forest Performance                                             | 39        |
| 4.2.1.2  | Graph Neural Network Performance                                      | 40        |
| 4.2.1.3  | GNN vs. Traditional ML Baselines                                      | 40        |
| 4.2.2    | Hyperparameter Tuning Results                                         | 41        |
| 4.3      | Nepali Medicinal Plant Virtual Screening Results                      | 42        |
| 4.3.1    | Screening Library Composition                                         | 42        |
| 4.3.2    | Multi-Model Screening Funnel and Hit Identification                   | 42        |
| 4.3.3    | Consensus Leads from the Best-Performing Models                       | 44        |
| 4.3.3.1  | Highest-Confidence Drug-Like Lead: Apigenin ( <i>Ocimum sanctum</i> ) | 46        |
| 4.3.3.2  | Top Consensus Lead: Luteolin                                          | 47        |
| 4.3.4    | Structural Diversity of Plant-Derived Leads                           | 47        |
| 4.4      | Discussion                                                            | 47        |
| 4.4.1    | Evaluation of the Research Hypotheses                                 | 47        |
| 4.4.2    | Performance of Graph Neural Networks Relative to Baselines            | 48        |
| 4.4.3    | Significance of Scaffold-Based Evaluation                             | 48        |
| 4.4.4    | Ethnopharmacological Rationale for Natural Product Screening          | 48        |

|          |                                                                        |           |
|----------|------------------------------------------------------------------------|-----------|
| 4.4.5    | Lead Compound Prioritisation Strategy . . . . .                        | 49        |
| 4.4.6    | Limitations of the Current Results . . . . .                           | 49        |
| <b>5</b> | <b>Conclusions and Recommendations</b>                                 | <b>51</b> |
| 5.1      | Conclusions . . . . .                                                  | 51        |
| 5.2      | Recommendations for Future Work . . . . .                              | 52        |
|          | <b>References</b>                                                      | <b>54</b> |
|          | <b>Appendices</b>                                                      |           |
|          | Appendix A: Dataset Information . . . . .                              | 57        |
|          | Appendix B: Screening Library and Virtual Screening Results . . . . .  | 59        |
|          | Appendix C: GNN Architecture and Hyperparameter Optimisation . . . . . | 61        |
|          | Appendix D: Software and Computational Environment . . . . .           | 63        |

# Chapter 1

## Introduction

### 1.1 Background

Cancer remains one of the most formidable public health challenges of the twenty-first century, imposing an extraordinary clinical, economic, and societal burden on populations worldwide. According to the GLOBOCAN 2022 estimates compiled by the International Agency for Research on Cancer (IARC), there were approximately 20 million new cancer cases and 9.7 million cancer-related deaths globally in 2022 (Siegel et al., 2023). Among all cancer subtypes, **lung cancer** is the leading cause of cancer mortality, accounting for approximately 1.8 million deaths and representing nearly 18% of all cancer fatalities in that year. It is the second most commonly diagnosed cancer by incidence and the most lethal, primarily because the majority of patients are diagnosed at advanced or metastatic stages when curative surgery is no longer feasible.

**Non-Small Cell Lung Cancer (NSCLC)** is the predominant histological category of lung cancer, constituting approximately 85% of all diagnoses. NSCLC encompasses three major subtypes: adenocarcinoma (~40%), squamous cell carcinoma (~25–30%), and large-cell carcinoma (~10–15%). The remaining 15% is classified as Small Cell Lung Cancer (SCLC), a neuroendocrine tumour with distinct biology and treatment approaches. The epidemiology of NSCLC has important geographical and ethnic dimensions: the disease burden is particularly acute across South and East Asia, including Nepal, where limited access to computed tomography (CT) screening programmes and low public awareness result in late-stage diagnosis in the majority of patients, precluding surgical intervention and significantly worsening prognosis (Rotow & Bivona, 2023). The five-year overall survival rate for stage IV NSCLC remains below 10% with conventional chemotherapy.

#### 1.1.1 The Epidermal Growth Factor Receptor in NSCLC

The **Epidermal Growth Factor Receptor (EGFR)**, also designated as ErbB1 or HER1, is a 134 kDa transmembrane receptor tyrosine kinase encoded by the *EGFR* gene on chromosome 7p11.2. It is a founding member of the ErbB receptor tyrosine kinase family, which comprises four closely related receptors: EGFR (ErbB1), HER2 (ErbB2), HER3 (ErbB3), and HER4 (ErbB4). Structurally, EGFR consists of an extracellular ligand-binding domain (domains I–IV), a single hydrophobic transmembrane helix, an intracellular juxtamembrane region, a bilobed tyrosine kinase catalytic domain, and a carboxy-terminal regulatory tail containing multiple autophosphorylation sites (Rotow & Bivona, 2023).

Under physiological conditions, EGFR is activated by the binding of cognate ligands, primarily Epidermal Growth Factor (EGF) and Transforming Growth Factor- $\alpha$  (TGF- $\alpha$ ). Ligand binding induces receptor homodimerisation or heterodimerisation with other ErbB family members, resulting in conformational rearrangement of the

intracellular kinase domain, release of autoinhibition, and trans-autophosphorylation of specific tyrosine residues (e.g., Tyr1068, Tyr1148, Tyr1173). These phosphorylated residues serve as docking sites for adaptor proteins and signalling effectors, initiating downstream cascades including:

- The **RAS-RAF-MEK-ERK** (MAPK) pathway, which promotes cell cycle progression, proliferation, and differentiation.
- The **PI3K-AKT-mTOR** pathway, which enhances cell survival, inhibits apoptosis, and regulates protein synthesis.
- The **STAT3/5** pathway, which drives transcription of pro-survival and immunomodulatory genes.
- The **PLC- $\gamma$**  pathway, which regulates intracellular calcium signalling and PKC activation.

In NSCLC, somatic activating mutations within the EGFR tyrosine kinase domain confer *ligand-independent* constitutive receptor activity, resulting in the continuous, dysregulated stimulation of these pro-survival pathways and the driving of oncogenesis. The two most clinically prevalent sensitising mutations account for 85–90% of all EGFR-mutant NSCLC:

1. **Exon 19 deletions (Ex19del)**: In-frame deletions of 9–24 nucleotides removing four conserved residues (LREA) from the C-helix of the kinase domain. These account for approximately 45–50% of EGFR-mutant NSCLC.
2. **L858R point mutation**: A leucine-to-arginine substitution at position 858 in the activation loop of the kinase domain, accounting for approximately 40% of EGFR-mutant NSCLC.

The prevalence of EGFR-activating mutations varies profoundly by ethnicity and geographic region. In Caucasian NSCLC patients, EGFR mutations are detected in 10–15% of cases, whereas in East Asian populations (including Chinese, Japanese, and Korean patients), the prevalence reaches 40–50% (Leonetti et al., 2019). This disparity has profound implications for Nepal and the broader South Asian region: given the rising NSCLC incidence and the genetically admixed population spanning South and East Asian ancestries, a significant and growing proportion of Nepali NSCLC patients harbour targetable EGFR mutations. However, access to molecular diagnostic testing and novel targeted therapies remains highly limited in Nepal, making computational drug discovery approaches particularly relevant for identifying affordable new candidate compounds.

### 1.1.2 Generations of EGFR Tyrosine Kinase Inhibitors

The clinical development of EGFR tyrosine kinase inhibitors (TKIs) has proceeded in three successive generations, each designed to overcome the resistance mechanisms that emerged against its predecessor.

**First-generation TKIs** (gefitinib, erlotinib) are reversible, competitive inhibitors of the EGFR ATP-binding pocket that bind preferentially to the mutant receptor over wild-type EGFR. Large randomised controlled trials, including IPASS (gefitinib) and EORTC (erlotinib), demonstrated unprecedented response rates of 55–75% and progression-free survival of 9–13 months in patients harbouring sensitising EGFR mutations, revolutionising the treatment of EGFR-mutant NSCLC. Despite these remarkable initial responses, acquired resistance emerged in virtually all patients, most commonly (in 50–60% of cases) through the **T790M gatekeeper mutation** in exon 20. This methionine substitution at Thr790 sterically occludes the binding of first-generation inhibitors and increases the binding affinity of the kinase for ATP, effectively out-competing reversible TKIs (Leonetti et al., 2019).

**Second-generation TKIs** (afatinib, dacomitinib) are irreversible, covalent inhibitors that form a Michael addition bond with Cys797 in the ATP-binding cleft, providing broader kinase inhibition and some activity against T790M-mutant EGFR in preclinical models. However, the irreversible inhibition of wild-type EGFR in skin, gut, and other normal tissues produces significant dose-limiting on-target toxicities (severe skin rash, diarrhoea, mucositis), which restrict clinical dosing to levels insufficient to overcome T790M resistance effectively.

**Third-generation TKIs** (osimertinib, lazertinib, rociletinib) were rationally designed to selectively target T790M-mutant EGFR through covalent binding to Cys797, while achieving greater selectivity for mutant over wild-type EGFR. Osimertinib (AstraZeneca, AZD9291), approved by the FDA in 2015, has become the definitive standard of care for EGFR-mutant NSCLC, demonstrating superior efficacy (PFS 18.9 months vs. 10.2 months for comparators), excellent CNS penetration (addressing brain metastases), and an improved toxicity profile compared to earlier generations (Rotow & Bivona, 2023).

**The C797S resistance problem.** Despite osimertinib’s superiority, acquired resistance emerges in virtually all patients within 9–18 months. The most common resistance mechanism is the **C797S substitution** at the covalent binding site (Cys797 → Ser797), which abrogates the irreversible covalent bond formation required for osimertinib activity. When C797S occurs in *cis* with T790M (on the same allele), the resulting **triple-mutant EGFR** forms (Ex19del/T790M/C797S and L858R/T790M/C797S) are *resistant to all three generations* of currently approved EGFR TKIs. Additional resistance mechanisms include MET amplification, HER2 amplification, RAS/RAF mutations, and SCLC histological transformation (Leonetti et al., 2019). As of 2026, there is no approved therapy for triple-mutant EGFR, and it remains a major long-term unmet need in NSCLC treatment. It is important, however, to state the scope of this thesis precisely at the outset: because no large-scale bioactivity dataset yet exists for the triple mutant, the present work develops and validates its computational framework on the abundant *aggregate* EGFR inhibition data available in ChEMBL (ChEMBL203)—which is dominated by wild-type and single-mutant measurements. The framework therefore targets *general* EGFR inhibitor identification; extending it to mutant-specific prediction, including the triple mutant, is a direction for future work rather than a claim of this study.

### 1.1.3 Computational Drug Discovery and Graph Neural Networks

Traditional drug discovery is a slow, costly, and attrition-prone process. The average time from target identification to regulatory approval is 10–15 years, and the average cost per approved drug has been estimated at approximately USD 2.0–2.6 billion (Schneider & Clark, 2020). The high attrition rate (~90% of compounds entering Phase I clinical trials fail) is driven primarily by inadequate efficacy, unexpected toxicity, and poor pharmacokinetic properties, all of which are increasingly addressable through computational prediction.

**Computer-Aided Drug Design (CADD)** encompasses a broad spectrum of computational techniques including molecular docking, molecular dynamics simulation, free energy perturbation, QSAR modelling, and virtual screening. Among these, machine learning-based virtual screening has emerged as the most scalable approach for the rapid prioritisation of large compound libraries (Walters & Murcko, 2020).

**Graph Neural Networks (GNNs)** represent a particularly powerful class of machine learning architectures for molecular property prediction (Gilmer et al., 2017). The fundamental insight is that molecules can be represented as mathematical graphs in which atoms correspond to nodes carrying feature vectors and chemical bonds correspond to edges carrying their own feature vectors, as illustrated in Figure 1. GNNs learn molecular representations through iterative message-passing: at each layer, every atom aggregates feature information from its bonded neighbours and updates its own hidden representation accordingly. After several such message-passing layers, a readout step aggregates all the atom representations into a single fixed-length molecular embedding that is passed to a classifier or regression head.

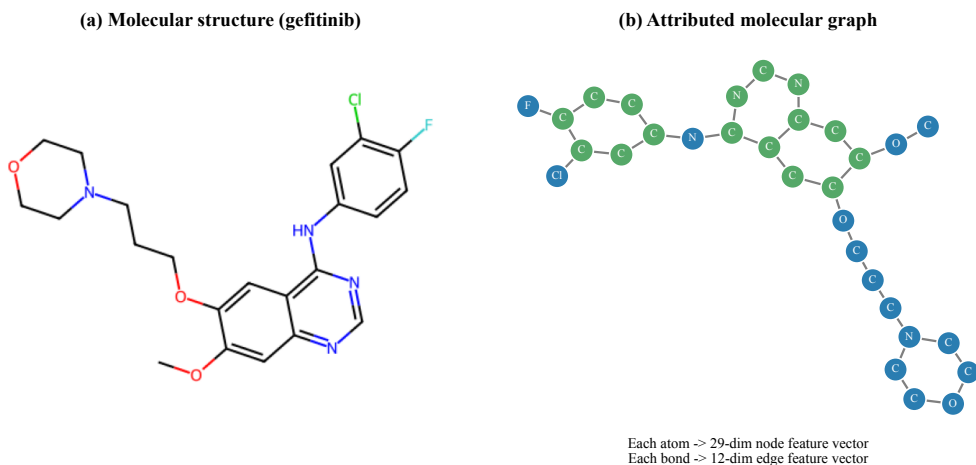


Figure 1: Representation of a molecule as an attributed graph for graph neural network learning. The 2D structure of the EGFR inhibitor gefitinib (a) is converted into a molecular graph (b) in which each atom becomes a node and each bond an edge. Aromatic atoms are shown in green and non-aromatic atoms in blue; each atom and bond is encoded by a fixed-length feature vector.

This approach offers several advantages over traditional fingerprint-based molecular representations. First, GNNs learn task-specific molecular representations *end-to-end* from bioactivity data, rather than relying on pre-defined structural patterns encoded in fingerprints. Second, by incorporating edge feature vectors, GNNs can natively encode bond-

level information (bond order, conjugation, stereochemistry) that is either discarded or only coarsely approximated in circular fingerprints. Third, deep GNN architectures can capture long-range atomic interactions spanning multiple bonds, enabling them to model higher-order pharmacophoric relationships that influence molecular activity (Jiang et al., 2021).

Despite these advantages, the translation of GNNs from academic benchmarks to practical drug discovery applications faces challenges, including the need for rigorous evaluation protocols, chemical interpretability, and extension to prospective screening pipelines—all of which are directly addressed by this thesis.

## 1.2 Problem Statement

Despite the transformative impact of EGFR-targeted therapy and the promise of Graph Neural Networks for drug discovery, critical and interrelated gaps persist in both clinical practice and computational methodology.

### 1.2.1 Clinical and Translational Need

Acquired resistance to EGFR tyrosine kinase inhibitors—most notably the **C797S mutation**, which renders third-generation covalent inhibitors ineffective and for which no fourth-generation drug is yet approved—sustains a continuing need to discover structurally novel, non-covalent EGFR inhibitor chemotypes (Rotow & Bivona, 2023). This need is most acute in South and East Asia, where EGFR-mutant NSCLC is highly prevalent yet novel targeted therapies remain expensive and computational drug-discovery infrastructure is scarce. There is therefore a clear demand for open-source, CPU-executable screening pipelines that can prioritise affordable, regionally accessible natural-product leads—the translational problem this thesis addresses.

### 1.2.2 Computational Problem: Methodological Gaps in GNN-Based EGFR Inhibitor Prediction

A review of the existing computational literature reveals three specific methodological deficiencies that this thesis is designed to address:

1. **Evaluation bias from random splitting.** The majority of published GNN models for molecular property prediction, including those applied to EGFR inhibitor classification, employ random train/test splitting. This approach allows structurally similar or near-identical molecular scaffolds to appear simultaneously in the training and test sets. As a result, the reported performance metrics substantially overestimate a model’s ability to generalise to genuinely novel chemical matter—precisely the task required in prospective virtual screening. A model that has “memorised” a scaffold from the training set cannot be trusted to correctly classify a structurally unprecedented compound in a real screening campaign.
2. **Limited molecular featurisation.** Many published GNN models for EGFR or related kinase targets use minimal node feature sets (e.g., atomic number only, or a small set of atom type flags) and ignore edge (bond) features entirely. This discards potentially discriminative chemical information including bond order,

conjugation, aromaticity, and stereochemistry—all of which influence molecular recognition in the EGFR binding pocket.

3. **Absence of prospective virtual screening on natural product libraries.** Most academic GNN studies in the EGFR domain report classification performance on held-out test sets but do not extend their trained models to the most practically relevant application: scoring and ranking an external library of unseen compounds to identify novel lead candidates—especially compounds derived from traditional medicinal plants that may offer affordable, regionally accessible therapeutic options (Malik et al., 2025).

This thesis directly addresses all three computational gaps through an end-to-end pipeline combining rigorous scaffold-aware evaluation, rich multi-dimensional graph featurisation, and prospective virtual screening of Nepali medicinal plant compounds. Because large-scale triple-mutant bioactivity data are not yet available, the classifier is trained on aggregate EGFR data, and the natural-product scaffolds it surfaces are intended as candidate starting points for future experimental validation rather than as validated triple-mutant inhibitors.

## 1.3 Objectives

### 1.3.1 General Objective

To develop and rigorously benchmark a Graph Neural Network framework for EGFR inhibitor classification, and to deploy the best-performing models for prospective virtual screening of Nepali medicinal plant compounds as candidate anti-NSCLC leads.

### 1.3.2 Specific Objectives

1. To build graph neural network models that classify EGFR inhibitors and compare them with traditional machine-learning models.
2. To use the best models to screen Nepali medicinal plant compounds and identify drug-like EGFR inhibitor leads.

## 1.4 Rationale

**Clinical rationale.** Acquired resistance to EGFR TKIs—and ultimately the triple-mutant EGFR that follows osimertinib failure—continues to drive the search for structurally novel inhibitor chemotypes, yet bringing new therapies through conventional drug discovery takes years to decades. Computational virtual screening can accelerate the early stage of this search by pre-selecting the most promising candidates from large chemical libraries for experimental testing, reducing the number of compounds that must be synthesised and assayed. In Nepal and South Asia, where EGFR mutations are prevalent yet expensive targeted therapies remain inaccessible, a screening pipeline built only from open-source tools and public databases is a practical and scalable contribution (Rotow & Bivona, 2023).

**Scientific rationale.** Graph Neural Networks have demonstrated strong performance across molecular machine learning benchmarks (Hu et al., 2020), yet their application to EGFR inhibitor discovery has been constrained by the three methodological shortcomings enumerated in Section 1.2. By addressing all three gaps simultaneously within a single, reproducible pipeline—scaffold splitting, rich featurisation, systematic multi-architecture comparison, and prospective Nepali plant compound screening—this thesis provides a methodologically complete framework that sets a higher standard for future GNN-based kinase inhibitor studies.

**Data science rationale.** This project integrates multiple core data science competencies within a single applied research context: large-scale database querying and API integration (ChEMBL), cheminformatics feature engineering (RDKit), deep learning architecture design and training (PyTorch, PyTorch Geometric), automated hyperparameter search (Optuna), prospective natural compound screening (COCONUT 2.0), and scientific visualisation (Matplotlib, Seaborn, RDKit 2D depictions). The project therefore constitutes a comprehensive demonstration of advanced data science methodology applied to a high-impact biomedical problem.

**Practical rationale.** The entire pipeline is implemented using exclusively open-source Python libraries (RDKit v2026.3.2, PyTorch Geometric v2.7.0, Optuna v4.8.0, scikit-learn v1.8.0) and publicly available databases (ChEMBL, COCONUT 2.0), all executing on commodity CPU hardware. This ensures full reproducibility, zero licensing cost, and broad accessibility for researchers in academic settings.

## 1.5 Research Hypotheses

The following two testable research hypotheses were formulated prior to experimentation and are evaluated against the results presented in Chapter 4:

1.  $H_1$  (**Performance Hypothesis**): Graph Neural Network architectures incorporating comprehensive multi-dimensional atomic node features and bond-level edge features, when optimised via Bayesian hyperparameter search, will achieve measurably superior scaffold test-set AUROC compared to traditional machine learning baselines (Random Forest and MLP on 2048-bit ECFP4 fingerprints) under a strict Bemis-Murcko scaffold split on the ChEMBL EGFR bioactivity dataset.
2.  $H_2$  (**Virtual Screening Hypothesis**): The virtual screening pipeline, employing the study’s best-performing classifiers to score compounds from seven traditional Nepali medicinal plant species retrieved from COCONUT 2.0, will identify at least one fully Lipinski-compliant natural-product-derived lead compound with a predicted activity probability  $\hat{p} \geq 0.80$  and a Quantitative Estimate of Drug-likeness (QED) score  $\geq 0.40$ , contributing a viable experimental lead candidate for affordable NSCLC therapy development in the South Asian context.

## 1.6 Significance of the Study

This research makes several original and complementary contributions to the intersection of computational data science and molecular drug discovery:

**Methodological contribution.** This thesis provides the first systematic, side-by-side comparison of five GNN architectures (GCN, GAT, GIN, AttentiveFP, GraphSAGE) with a rich 29-dimensional atomic featurisation scheme and 12-dimensional bond feature encoding, evaluated under a rigorous Bemis-Murcko scaffold split for the binary classification of EGFR inhibitor bioactivity. The use of Bayesian hyperparameter optimisation (Optuna) provides a reproducible and statistically principled model selection procedure.

**Natural product screening contribution.** By curating bioactive compounds from seven traditional Nepali medicinal plant species (COCONUT 2.0) and applying the study’s best-performing models (the best graph network and the best overall classifier) for prospective EGFR inhibition prediction, this thesis delivers the first computational evaluation of Nepali natural products as potential EGFR TKI candidates—directly relevant to the South Asian oncology community where EGFR mutations are prevalent and novel therapies remain expensive.

**Translational contribution.** By extending the best-performing models into a complete multi-model prospective virtual screening pipeline applied to Nepali medicinal plant compounds and generating a ranked list of Lipinski-compliant natural-product-derived lead candidates with computed ADMET-proxy properties, the thesis delivers a concrete, actionable output—a prioritised shortlist of natural compounds for future experimental validation—rather than a purely academic performance comparison.

**Regional and social contribution.** The elevated prevalence of EGFR-activating mutations in East and South Asian NSCLC patients, combined with the rising cancer burden and limited access to novel therapies in Nepal, makes this research particularly relevant to the South Asian academic and clinical community. The entirely open-source, CPU-executable implementation ensures that the methodology can be adopted and extended by researchers in resource-limited settings without access to expensive commercial software or high-performance computing clusters.

## 1.7 Limitations

The following limitations of the current study are acknowledged and should be considered when interpreting the results:

- **No experimental validation.** All compound classifications and virtual screening predictions are purely computational. No *in vitro* biochemical assay, cell-based proliferation assay, or *in vivo* pharmacological experiment was conducted within the scope of this thesis. The predicted lead compounds remain computational hypotheses requiring experimental confirmation before any biological conclusions can be drawn.
- **Bioassay data heterogeneity.** Bioactivity data sourced from ChEMBL aggregates IC<sub>50</sub> measurements originating from hundreds of independent laboratories employing diverse assay formats, cell lines, and experimental conditions. This inherent variability introduces label noise into the training dataset. While pIC<sub>50</sub> averaging and strict data filtering mitigate this to some extent, residual heterogeneity may impose a ceiling on achievable test-set performance under scaffold splitting.

- **Wild-type EGFR specificity.** No dedicated, large-scale bioactivity dataset exists for triple-mutant EGFR (Ex19del/T790M/C797S) in ChEMBL. Consequently, the model was trained on the aggregate EGFR bioactivity data of ChEMBL203, which is dominated by wild-type and single-mutant measurements. While the prospective screening of structurally novel Nepali plant scaffolds is designed to surface chemotypes distinct from existing covalent inhibitors, the model cannot explicitly predict activity against the C797S mutant form.
- **Incomplete database coverage of Nepali medicinal flora.** The virtual screening library is limited to compounds recorded for the seven target species in the COCONUT 2.0 database. Coverage is uneven across species and many traditional Nepali plant compounds remain uncharacterised and absent from public cheminformatics databases, so the screened library is not exhaustive of the phytochemical diversity of these plants.
- **Model-sensitivity of the virtual screen.** The classifiers were trained on synthetic kinase inhibitors but applied to structurally dissimilar natural products, and the resulting hit list depends strongly on the model. As reported in Chapter 4, the number of predicted actives and the identity of the top leads differ markedly between the tuned AttentiveFP, the best graph network (GCN), and the best classifier (Random Forest). No single model can be treated as authoritative; the study mitigates this by deploying the two best models and prioritising cross-model consensus leads, but only experimental assay can resolve the residual uncertainty.
- **Out-of-distribution prediction reliability.** On chemical space far from the training distribution, the weakest model (AttentiveFP) assigned near-saturated probabilities to large polyphenolic tannins that fail drug-likeness filters, and even the best models still ranked some large, non-drug-like polyphenols and triterpenoids highly—indicating that such predictions must be interpreted with caution and coupled with physicochemical filtering.
- **Two-dimensional molecular representation.** The molecular graph encodes the constitutional (2D) topology of the molecule: atom identity and bonding connectivity. Three-dimensional conformational effects, induced-fit protein-ligand binding geometry, explicit solvent interactions, and protein-mediated allosteric effects are not represented in the input features.
- **Binary classification scope.** The study frames EGFR inhibition as a binary classification problem. This formulation discards the continuous nature of the  $\text{pIC}_{50}$  scale and cannot distinguish between a moderately potent compound ( $\text{pIC}_{50} = 6.1$ ) and a highly potent compound ( $\text{pIC}_{50} = 9.5$ ). Continuous regression modelling of  $\text{pIC}_{50}$  values is beyond the current scope but is recommended for future work.
- **ADMET-proxy limitations.** The ADMET profiling in this study relies solely on RDKit-computed physicochemical descriptors (MW, LogP, HBD, HBA, TPSA, QED) as proxies for oral bioavailability. These filters do not capture metabolic stability, cytochrome P450 inhibition, plasma protein binding, blood-brain barrier permeability, hERG channel toxicity, or other critical ADMET parameters that determine clinical development success.

## 1.8 Organisation of the Thesis

The remainder of this thesis is organised as follows. **Chapter 2** provides a comprehensive review of the relevant literature, covering the biology of EGFR and NSCLC, the clinical development and resistance mechanisms of EGFR TKIs, public bioactivity databases, molecular representation methods, GNN architectures for drug discovery, traditional Nepali medicinal plants as a source of EGFR inhibitor candidates, and virtual screening approaches. **Chapter 3** describes the complete materials and methods of the five-phase computational pipeline, including data acquisition, molecular graph construction, model architectures, training protocol, hyperparameter optimisation, and the Nepali medicinal plant virtual screening workflow. **Chapter 4** presents all experimental results with tables and analytical interpretation, covering dataset characterisation, model benchmarking, hyperparameter tuning outcomes, and Nepali plant virtual screening lead prioritisation. **Chapter 5** draws the principal conclusions of the thesis and recommends directions for future experimental and computational work.

## Chapter 2

### Literature Review

This chapter provides a comprehensive review of the scientific literature underpinning this thesis. Section 2.1 reviews the molecular biology of EGFR and its role in NSCLC oncogenesis. Section 2.2 discusses the three generations of EGFR TKIs and their clinical limitations. Section 2.3 describes the ChEMBL bioactivity database and associated data quality challenges. Section 2.4 reviews molecular representation methods for machine learning. Section 2.5 surveys the GNN architectures employed in this study. Section 2.6 reviews the Nepali medicinal plants and natural product databases used for virtual screening. Section 2.7 reviews virtual screening methodologies. Section 2.8 synthesises the identified research gaps that motivate this thesis.

#### 2.1 EGFR Kinase Biology and NSCLC

##### 2.1.1 Molecular Architecture of EGFR

The Epidermal Growth Factor Receptor (EGFR), also designated ErbB1 or HER1, is a 134 kDa single-pass transmembrane receptor tyrosine kinase (RTK) encoded by the *EGFR* gene on chromosome 7p11.2. EGFR is the founding member of the ErbB receptor family (ErbB1–ErbB4), which collectively orchestrates fundamental cellular processes including proliferation, differentiation, survival, and motility (Hynes & Lane, 2005). Structurally, EGFR is organised into four functional domains:

1. **Extracellular domain** (domains I–IV): A glycosylated ectodomain that binds EGF-family ligands (EGF, TGF- $\alpha$ , amphiregulin, epiregulin, betacellulin, and others) through domains I and III, while domain II mediates receptor dimerisation.
2. **Transmembrane domain**: A single  $\alpha$ -helical segment that anchors the receptor in the plasma membrane and transmits conformational changes from the extracellular to the intracellular compartment.
3. **Juxtamembrane domain**: An intracellular regulatory segment that modulates kinase activity through direct interaction with the kinase domain.
4. **Intracellular kinase domain**: A bilobed catalytic domain (N-lobe and C-lobe) that binds ATP and catalyses phosphotransfer to substrate tyrosine residues. The ATP-binding cleft lies at the interface of the two lobes and contains the hinge region, the P-loop, the activation loop, and the DFG motif. The kinase domain is followed by a carboxy-terminal tail bearing multiple autophosphorylation sites (Tyr992, Tyr1045, Tyr1068, Tyr1086, Tyr1148, Tyr1173) that recruit downstream effector proteins.

Under normal physiological conditions, EGFR is maintained in an autoinhibited, inactive state characterised by an asymmetric kinase dimer configuration. Ligand binding induces receptor homo- or heterodimerisation with other ErbB family members (HER2, HER3, HER4), leading to relief of autoinhibition, *trans*-autophosphorylation of key tyrosine residues on the carboxy-terminal tail, and recruitment of SH2- and PTB-domain-containing adaptor proteins (Siegel et al., 2023). These phosphorylated docking sites initiate multiple downstream signalling cascades:

- The **RAS-RAF-MEK-ERK (MAPK) pathway**: Activated by recruitment of GRB2-SOS complex, driving G1-to-S cell cycle progression, cell proliferation, and differentiation.
- The **PI3K-AKT-mTOR pathway**: Activated via p85/p110 PI3K recruitment, promoting cell survival, inhibiting apoptosis (via BAD phosphorylation and FOXO suppression), and stimulating protein synthesis.
- The **STAT3/5 signalling pathway**: Direct EGFR kinase-mediated phosphorylation of STAT3 and STAT5, driving transcription of anti-apoptotic (Bcl-xL) and proliferative (cyclin D1) genes.
- The **PLC- $\gamma$  pathway**: Activation of phospholipase C- $\gamma$ , generating IP3 and diacylglycerol, which mobilise intracellular calcium and activate PKC isoforms.

### 2.1.2 Oncogenic EGFR Mutations in NSCLC

In non-small cell lung cancer (NSCLC), somatic gain-of-function mutations within the EGFR kinase domain render the receptor constitutively active in a ligand-independent manner, producing continuous, dysregulated stimulation of the pro-survival and proliferative pathways described above (Rotow & Bivona, 2023). This oncogenic signalling drives uncontrolled tumour growth, resistance to apoptosis, angiogenesis, and metastatic dissemination.

The two most clinically prevalent sensitising EGFR mutations, which together account for 85–90% of all EGFR-mutant NSCLC, are:

- **Exon 19 deletions (Ex19del)**: A heterogeneous group of in-frame deletions of 9–24 base pairs within exon 19, most commonly removing the LREA motif (residues 747–750) from the  $\alpha$ C-helix of the kinase N-lobe. These deletions stabilise the active conformation of the kinase domain by repositioning the  $\alpha$ C-helix, reducing the activation energy for catalysis. Ex19del accounts for approximately 45–50% of EGFR-mutant NSCLC.
- **L858R point mutation**: A leucine-to-arginine substitution at position 858 within the activation loop (A-loop) of the kinase domain. This mutation disrupts the autoinhibitory network involving Leu858, Asp855, and Phe856 of the DFG motif, shifting the equilibrium from the inactive to the active kinase conformation. L858R accounts for approximately 40% of EGFR-mutant NSCLC.

Rare sensitising mutations (<5% combined) include exon 18 G719X, exon 20 S768I, and exon 21 L861Q. These display differential sensitivity to approved TKIs and often require alternative treatment strategies.

The prevalence of EGFR mutations in NSCLC exhibits a profound ethnic disparity. In Caucasian populations, EGFR mutations are found in approximately 10–15% of NSCLC cases, whereas in East Asian populations (Chinese, Japanese, Korean, and related groups), the prevalence reaches 40–50% (Leonetti et al., 2019). This epidemiological pattern reflects underlying differences in the frequency of germline polymorphisms at the *EGFR* locus that predispose to somatic mutation acquisition. Importantly, this disparity has direct clinical relevance for South and East Asia, including Nepal, where EGFR-targeted therapy represents the most impactful precision oncology intervention available, yet access to diagnostic testing and novel therapeutic agents remains significantly constrained.

## 2.2 Three Generations of EGFR TKIs and Resistance Mechanisms

### 2.2.1 First-Generation Reversible TKIs

The clinical development of EGFR tyrosine kinase inhibitors began with the regulatory approval of gefitinib (Iressa, AstraZeneca) in 2003 and erlotinib (Tarceva, Genentech/Roche) in 2004. Both agents are reversible, ATP-competitive small molecule inhibitors that occupy the EGFR ATP-binding cleft through a combination of a hinge-region hydrogen bond (to the backbone amide nitrogen of Met793), hydrophobic contacts with the ribose pocket, and van der Waals interactions with the hydrophobic regions I and II of the binding site. Crystal structures of EGFR bound to gefitinib (PDB: 2ITY) and erlotinib (PDB: 1M17) have confirmed these binding modes at the atomic level (Stamos et al., 2002).

Landmark clinical trials established the superiority of first-generation TKIs over platinum-doublet chemotherapy in patients with sensitising EGFR mutations. The IPASS trial (gefitinib vs. carboplatin/paclitaxel) and EURTAC trial (erlotinib vs. platinum doublet) demonstrated objective response rates of 55–75% and median progression-free survival (PFS) of 9–13 months—unprecedented outcomes for a molecularly unselected metastatic NSCLC population.

However, acquired resistance develops in virtually all patients within 9–13 months. The most common resistance mechanism (50–60% of cases) is the **T790M gatekeeper mutation** (Thr790Met) in exon 20 of the EGFR kinase domain (Leonetti et al., 2019). This methionine substitution acts through two complementary mechanisms: (i) steric clash with the quinazoline scaffold of first-generation inhibitors due to the bulkier methionine side chain, and (ii) a 100-fold increase in the binding affinity of the kinase for ATP, effectively out-competing the reversible TKI. Additional resistance mechanisms include MET amplification (~5–15%), HER2 amplification (~5–10%), PIK3CA mutation (~5%), and histological transformation to small cell carcinoma (~5%) (Leonetti et al., 2019).

### 2.2.2 Second-Generation Irreversible TKIs

The T790M-mediated resistance to first-generation TKIs motivated the development of second-generation irreversible inhibitors: afatinib (Gilotrif, Boehringer Ingelheim) and dacomitinib (Vizimpro, Pfizer). These agents form a covalent Michael addition bond with the thiol group of Cys797 at the rim of the ATP-binding cleft, providing kinetically

stable, irreversible target engagement. As pan-ErbB inhibitors, they also inhibit HER2 and HER4 kinase activity.

Afatinib demonstrated superiority over platinum-doublet chemotherapy in the LUX-Lung 3 and LUX-Lung 6 trials, and superior PFS versus gefitinib in LUX-Lung 7, establishing it as a first-line option for EGFR-mutant NSCLC. Dacomitinib similarly improved PFS over gefitinib in the ARCHER 1050 trial. However, the irreversible pan-ErbB inhibition profile of second-generation agents produces significant dose-limiting on-target toxicity—primarily grade 3/4 diarrhoea, skin rash, stomatitis, and paronychia—arising from inhibition of wild-type EGFR in the gastrointestinal epithelium and skin (Leonetti et al., 2019). This toxicity profile restricts clinical dosing to sub-therapeutic levels that are insufficient to overcome T790M-driven resistance in the majority of patients.

### 2.2.3 Third-Generation Covalent TKIs and the C797S Resistance Problem

The clinical unmet need created by T790M resistance drove the development of third-generation mutant-selective covalent TKIs, led by osimertinib (Tagrisso, AstraZeneca; AZD9291). Osimertinib was rationally designed through structure-based drug design to selectively engage T790M-mutant EGFR while sparing wild-type EGFR. Its indole-based acrylamide scaffold binds the EGFR ATP cleft and forms an irreversible covalent bond with Cys797 through a Michael addition, while its selectivity for mutant over wild-type EGFR is achieved through differential binding geometries and kinetics rather than steric exclusion (Rotow & Bivona, 2023).

The AURA3 trial established osimertinib as the standard second-line therapy for T790M-positive NSCLC after first-generation TKI failure. Subsequently, the FLAURA trial demonstrated osimertinib's superiority over first-generation TKIs as first-line therapy (median PFS 18.9 vs. 10.2 months; overall survival 38.6 vs. 31.8 months), establishing it as the unambiguous standard of care for EGFR-mutant NSCLC regardless of prior treatment. Additional advantages include excellent CNS penetration (~75% brain metastasis response rate) and an improved safety profile compared to afatinib.

Despite these remarkable clinical achievements, acquired resistance to osimertinib emerges in virtually all patients within 9–18 months (Leonetti et al., 2019). The dominant resistance mechanism is the **C797S mutation** (Cys797Ser), which substitutes the cysteine residue required for covalent TKI binding with a serine, abolishing the Michael addition reaction. C797S is detected in 20–40% of post-osimertinib biopsies and is of particular clinical importance when it occurs in *cis* with T790M on the same allele, generating the **triple-mutant EGFR** (Ex19del/T790M/C797S and L858R/T790M/C797S) that is resistant to *all three generations* of approved EGFR TKIs:

- First-generation TKIs cannot overcome T790M.
- Second-generation TKIs cannot overcome T790M steric clash at therapeutic doses.
- Third-generation TKIs cannot bind due to loss of the Cys797 covalent anchor.

Additional post-osimertinib resistance mechanisms include acquired MET amplification (~15%), HER2 amplification (~5%), RAS/RAF/MAP2K mutations (~10%), EGFR

amplification, EGFR C797 allosteric mutations (e.g., G796S, L792 mutations), and histological transformation to SCLC or squamous cell carcinoma (Rotow & Bivona, 2023). As of 2026, no approved therapeutic agent effectively addresses triple-mutant EGFR, making the discovery of fourth-generation EGFR inhibitors or non-covalent EGFR inhibitors the highest clinical priority in NSCLC drug development.

## 2.3 Bioactivity Databases and the ChEMBL Database

### 2.3.1 Overview of the ChEMBL Database

The ChEMBL database, maintained by the European Molecular Biology Laboratory-European Bioinformatics Institute (EMBL-EBI), is the largest and most extensively curated public repository of bioactive small molecules and their quantitative biological activities (Mendez et al., 2019). As of recent releases, the database contains approximately:

- >2.37 million distinct bioactive compounds
- >20.7 million bioactivity measurements
- >14,000 therapeutic targets from multiple organisms
- >85,000 document sources (primarily peer-reviewed publications)

ChEMBL is constructed primarily through systematic curation of bioactivity data from published medicinal chemistry and pharmacology literature, supplemented by direct data depositions from industry partners, academic consortia (e.g., CanSAR, SMACC), and high-throughput screening datasets (e.g., PubChem BioAssay, GSK PKIS). The EGFR kinase entry (target ID: ChEMBL203) is one of the most extensively annotated targets in the database, reflecting decades of intense drug discovery activity around this oncology target.

### 2.3.2 Bioactivity Endpoints and $\text{pIC}_{50}$ Conversion

The primary bioactivity endpoint used in this study is the **half-maximal inhibitory concentration** ( $\text{IC}_{50}$ ), defined as the concentration of a compound required to inhibit the target activity by 50% under specified assay conditions.  $\text{IC}_{50}$  values are expressed in molar concentration units (typically nanomolar, nM) and span many orders of magnitude across structurally diverse compounds.

For machine learning purposes,  $\text{IC}_{50}$  values are routinely converted to the logarithmic  $\text{pIC}_{50}$  scale by taking the negative base-ten logarithm of the molar concentration, so that higher  $\text{pIC}_{50}$  values correspond to more potent compounds. This transformation compresses the large dynamic range of  $\text{IC}_{50}$  values (spanning 6–8 orders of magnitude) into a more tractable numerical range (approximately 4–11 for drug-like compounds), improves the normality of the activity distribution, and is the standard practice in QSAR modelling. A threshold of  $\text{pIC}_{50} \geq 6.0$  (corresponding to an  $\text{IC}_{50}$  of  $\leq 1 \mu\text{M}$ ) is the conventional medicinal chemistry criterion for classifying a compound as a “hit” or “active” in kinase inhibitor discovery (Wu et al., 2018).

### 2.3.3 Data Quality Challenges

Utilising ChEMBL data for machine learning presents several important quality challenges that must be addressed during data curation (Wu et al., 2018):

- **Assay heterogeneity:** IC<sub>50</sub> measurements originate from hundreds of independent laboratories employing different assay formats (biochemical kinase assay, cell-based viability assay, radiometric assay, TR-FRET assay), cell lines, substrate concentrations, ATP concentrations, temperature, and pH. These variations can shift absolute IC<sub>50</sub> values by 1–2 orders of magnitude, introducing systematic noise into the training labels.
- **Censored measurements:** A significant proportion of reported values are censored (greater-than or less-than, e.g., IC<sub>50</sub> > 10,000 nM or IC<sub>50</sub> < 0.1 nM), indicating that the compound’s activity exceeded the assay’s detection limits. Standard practice restricts training data to exact (“=”) measurements only.
- **Replicate measurements:** The same compound may have been tested multiple times across different publications. Aggregating multiple pIC<sub>50</sub> measurements using the median is the approach adopted in this study to reduce the influence of outliers.
- **Structural redundancy:** Different chemical databases may use different SMILES notation conventions for the same compound (e.g., different salt forms, stereochemistry assignments). Canonicalisation using RDKit and salt stripping (retaining the largest fragment) are essential preprocessing steps to remove duplicates.
- **Activity cliff sensitivity:** Pairs of structurally similar compounds with dramatically different activity values (activity cliffs) create challenging regions of chemical space for both fingerprint-based and graph-based models.

## 2.4 Molecular Representation for Machine Learning

### 2.4.1 Hand-Crafted Fingerprint Descriptors

Molecular fingerprints are fixed-length binary or integer vectors that encode the presence or count of predefined structural features within a molecule. They represent the dominant molecular representation paradigm in traditional cheminformatics and remain strong baselines for bioactivity prediction tasks.

#### 2.4.1.1 Extended Connectivity Fingerprints (ECFP)

The Extended Connectivity Fingerprint (ECFP) (Rogers & Hahn, 2010), also known as the Morgan fingerprint (after its algorithmic basis in the Morgan algorithm for canonical atom numbering), is the most widely used molecular fingerprint in cheminformatics. ECFP encodes circular atomic neighbourhoods through an iterative procedure:

1. Each atom is assigned an initial integer identifier based on six atomic invariants (atomic number, degree, number of hydrogens, charge, isotope, and ring membership).
2. At each iteration  $r$  (radius), each atom’s identifier is updated by hashing together its current identifier with the identifiers of its immediate neighbours (corresponding to atoms within  $r$  bonds).
3. All identifiers generated at each iteration up to radius  $r$  are collected and folded into a bit vector of specified length (typically 1024 or 2048 bits) using a hash function.

ECFP4 (radius = 2, 2048 bits) is the de facto standard, encoding circular substructures up to 4 bonds in diameter. Its advantages include computational efficiency ( $O(N)$  in molecular size), interpretability of individual bits as specific substructural features, and strong empirical performance across diverse QSAR benchmarks. Limitations include: (i) fixed-length encoding discards variable-size topological patterns not captured within the predefined radius; (ii) bit collisions in folded fingerprints reduce information content; (iii) bond-level information (bond type, conjugation, stereochemistry) is only coarsely encoded; and (iv) the representation is not differentiable, precluding end-to-end learning of task-optimal molecular representations.

#### 2.4.1.2 Other Descriptor Types

Beyond fingerprints, molecular property prediction has traditionally relied on physico-chemical descriptors (e.g., RDKit 2D and 3D descriptors, MACCS keys, DRAGON descriptors), shape-based 3D pharmacophore fingerprints, and continuous molecular embeddings derived from SMILES string language models.

For the purposes of this study, ECFP4 fingerprints are used for both traditional machine learning baselines. **Random Forest (RF)** is an ensemble of 500 decision trees with balanced class weighting and `max_features='sqrt'`, providing a well-calibrated, sample-efficient baseline for medium-sized bioactivity datasets. The **Multi-Layer Perceptron (MLP)** baseline is a three-layer neural network (2048→512→128→1) operating on the same ECFP4 inputs, trained with BCEWithLogitsLoss and early stopping. The MLP bridges the gap between fingerprint-based and graph-based deep learning, allowing direct attribution of performance differences to the graph representation rather than to deep learning capacity in general.

#### 2.4.2 Graph-Based Molecular Representations

The fundamental insight motivating the graph-based approach is that a molecule is inherently a labelled graph: atoms are nodes with associated chemical properties, and bonds are edges with associated chemical characteristics. This isomorphism allows molecules to be fed directly into graph processing algorithms without information-losing pre-encoding steps (Gilmer et al., 2017).

A molecular graph is defined by three components:

- A set of **nodes**, where each node represents one heavy atom and carries a node feature vector describing that atom’s chemical properties.

- A set of **edges**, where each edge represents one chemical bond and carries an edge feature vector describing that bond’s properties.
- The **bonding connectivity**, recording which atoms are bonded to which.

The dimensionality and content of the node and edge feature vectors directly determine the chemical information available to the model. In this work, a comprehensive 29-dimensional node feature vector and 12-dimensional edge feature vector were designed to capture the chemical information most relevant to kinase inhibitor classification, as detailed in Chapter 3.

### 2.4.3 Graph Neural Networks: The Message-Passing Framework

The general message-passing neural network (MPNN) framework (Gilmer et al., 2017) provides a unified description of most GNN architectures. In this framework, each atom maintains a hidden representation that is refined layer by layer through two operations, illustrated in Figure 2:

**Aggregation:** each atom collects “messages” from its bonded neighbours, where each message is a learned function of the central atom’s representation, the neighbour’s representation, and the connecting bond features. The messages are then combined using a permutation-invariant operator (sum, mean, or max) so that the result does not depend on the arbitrary ordering of neighbours.

**Update:** each atom updates its own hidden representation by combining its previous state with the aggregated message, typically through a small multilayer perceptron or a gated recurrent unit.

After several message-passing layers, a **readout function** aggregates all the atom-level representations into a single graph-level embedding, which is then passed to a task-specific head (for example, a fully connected classifier for binary activity prediction).

The key advantages of the MPNN framework for molecular property prediction are: (i) the representation is learned end-to-end from bioactivity data, requiring no expert-designed feature engineering; (ii) arbitrary atom- and bond-level information can be incorporated via the node and edge feature vectors; (iii) the framework naturally handles molecules of varying size without fixed-length encoding; and (iv) the learned representations capture both local chemical environments (from lower layers) and global molecular topology (from higher layers through the readout).

## 2.5 Graph Neural Network Architectures for Drug Discovery

### 2.5.1 Graph Convolutional Network (GCN)

The Graph Convolutional Network (GCN) (Kipf & Welling, 2017) introduced the concept of spectral graph convolution through a computationally tractable first-order approximation. At each layer, every atom’s representation is updated by taking a degree-normalised average of the (linearly transformed) representations of its bonded neighbours together with its own, followed by a non-linear activation. The normalisation uses the atom degrees (with self-loops added) so that atoms with many bonds do not dominate the aggregation.

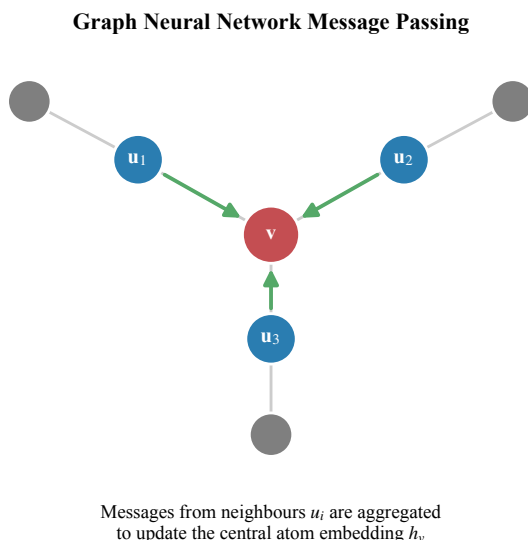


Figure 2: One step of graph neural network message passing. The central atom  $v$  updates its representation by aggregating learned messages from its bonded neighbours  $u_1, u_2, u_3$ . Repeating this over several layers lets each atom’s embedding incorporate progressively larger chemical neighbourhoods.

The GCN architecture has several notable limitations in the context of molecular property prediction. First, the degree-normalised aggregation weights all neighbouring atoms equally, without learning which chemical neighbours contribute most to a given property. Second, standard GCN implementations use node features only and do not incorporate edge features, discarding bond-level chemical information. Third, GCN is limited in its expressive power: it cannot distinguish certain pairs of non-isomorphic graph structures that differ only in their higher-order topology, limiting its ability to discriminate complex pharmacophoric patterns (Xu et al., 2019). Despite these limitations, GCN provides a strong and computationally efficient baseline and demonstrated competitive AUROC performance in the TDC ADMET benchmarks (Huang et al., 2021).

### 2.5.2 Graph Attention Network (GAT)

The Graph Attention Network (GAT) (Veličković et al., 2018) addresses the uniform weighting limitation of GCN by introducing learnable, data-driven attention coefficients for each pair of connected atoms. For each atom, the GAT computes a normalised attention weight for every neighbour from a learnable scoring function applied to the two atoms’ transformed representations. These weights, which sum to one across the neighbourhood, then determine how strongly each neighbour contributes to the atom’s updated representation, replacing the fixed degree-based averaging of the GCN with a learned, data-driven weighting.

To stabilise learning and capture diverse chemical interaction patterns, several parallel attention “heads” are typically used, with their outputs concatenated in intermediate layers and averaged in the final layer.

The GATv2 extension (Brody et al., 2022) resolves an expressivity limitation of the original GAT formulation (the “static attention” problem), in which the ordering of attention coefficients is query-independent. GATv2 modifies the attention scoring function to ensure that attention weights are dynamically conditioned on both the query and key node representations.

The attention mechanism of GAT is of particular value in molecular property prediction because different atoms contribute unequally to a given chemical property: in EGFR inhibitors, for example, the hinge-binding nitrogen atoms of the quinazoline pharmacophore are far more critical to EGFR activity than peripheral solvent-exposed substituents. GAT’s ability to learn and apply differential attention across atom pairs enables it to focus on pharmacophore-relevant atoms during representation learning.

### 2.5.3 Graph Isomorphism Network (GIN)

The Graph Isomorphism Network (GIN) (Xu et al., 2019) was developed from a theoretical analysis of the representational limits of GNNs within the message-passing framework. Xu et al. proved that the expressive power of any message-passing GNN is bounded by the Weisfeiler-Leman (WL) graph isomorphism test, and demonstrated that GIN achieves this theoretical upper bound. The GIN layer updates each atom by *summing* its neighbours’ representations, adding the atom’s own representation (scaled by a small learnable factor), and passing the result through a multilayer perceptron.

The critical theoretical insight is that sum aggregation preserves multiset information about neighbour features (unlike mean or max aggregation, which can conflate distinct neighbourhood configurations), making GIN maximally expressive among message-passing GNNs.

The GINEConv (Hu et al., 2020) extension incorporates edge features by adding a transformed bond feature vector to each neighbour’s representation before the summation, so that bond-level chemistry informs the aggregation.

Despite its theoretical superiority, GIN does not always outperform architecturally simpler GNNs on molecular property prediction benchmarks, suggesting that the WL-level expressivity gains are primarily relevant for highly symmetric graph structures that are uncommon in drug-like molecules. Empirically, GIN has shown strong performance on graph classification tasks in the TU graph benchmarks and MoleculeNet (Wu et al., 2018).

### 2.5.4 AttentiveFP

AttentiveFP (Xiong et al., 2020) is a graph neural network architecture purpose-built for molecular property prediction, distinguishing itself from generic GNN architectures through two key innovations: **atom-level attention** and **molecule-level iterative read-out**.

#### 2.5.4.1 Atom-Level Attention

At each message-passing step, AttentiveFP computes context-dependent atom-level attention weights that modulate the contribution of each neighbour to the central atom’s representation update. For each atom and neighbour pair, an attention score is computed

from the two atoms’ representations *together with the connecting bond’s feature vector*, and these scores are normalised across the neighbourhood to obtain the final attention weights.

This attention formulation is distinguished by its *native integration of edge features* in the attention computation: the bond feature vector is combined with the atom representations before the attention scoring step, allowing the model to condition attention weights on bond-level chemical information (for example, assigning higher attention to double bonds in aromatic systems).

#### 2.5.4.2 Molecule-Level Iterative Readout

After atom-level message passing, AttentiveFP employs a novel iterative readout procedure that refines a single molecule-level embedding over several timesteps. A context vector is initialised from the mean of all the final atom representations. At each timestep, the context vector attends over all atoms (computing normalised attention weights from the context and each atom representation), forms a weighted summary of the atoms, and uses a Gated Recurrent Unit (GRU) to update itself from that summary.

The use of a GRU enables the molecular embedding to be iteratively refined, allowing the model to capture increasingly global structural context with each timestep. The final context vector serves as the molecular embedding that is passed to the classification head.

AttentiveFP has been shown to achieve state-of-the-art or near-state-of-the-art performance across multiple MoleculeNet benchmarks (Jiang et al., 2021; Xiong et al., 2020), with the multi-step molecular attention mechanism and edge feature integration providing particularly significant advantages on property prediction tasks where long-range intramolecular interactions and bond-level chemistry are important. On this basis AttentiveFP was selected as the architecture on which this study focused its hyperparameter optimisation, although the benchmark in Chapter 4 ultimately found it to be the weakest of the five graph networks on the scaffold test set.

#### 2.5.5 GraphSAGE

GraphSAGE (Graph SAmple and aggreGatE) (Hamilton et al., 2017) is an inductive representation learning framework designed for graphs where new nodes must be embedded without retraining. Unlike GCN and GAT, which operate on the full graph adjacency matrix, GraphSAGE learns aggregation functions from *sampled* local neighbourhoods, making it particularly suitable for large-scale graphs.

The GraphSAGE update proceeds in two steps. First, each atom computes the mean of its neighbours’ representations to form a neighbourhood summary. Second, this summary is concatenated with the atom’s own previous representation and passed through a learnable transformation followed by a non-linear activation. The concatenation of the atom’s own representation with its neighbourhood aggregate ensures that the atom’s individual chemical identity is preserved alongside its local chemical context.

GraphSAGE offers two advantages in the molecular property prediction context: (i) mean aggregation is robust to variable-degree atoms without the normalisation instability that can affect degree-normalised GCN in molecules with high-degree atoms (e.g.,

bridgehead atoms in polycyclic scaffolds); and (ii) the inductive framework generalises well to novel scaffolds not seen at training time, which is directly relevant under Bemis-Murcko scaffold evaluation. In this study, GraphSAGE uses mean aggregation with BatchNorm and concatenated mean+max global pooling, following the same readout strategy as GCN.

## 2.6 Nepali Medicinal Plants and Natural Product Databases

### 2.6.1 Traditional Medicinal Plants in Nepal

Nepal harbours remarkable biodiversity, hosting approximately 6,500 flowering plant species within a geographic transect spanning tropical lowlands (~60 m a.s.l.) to alpine zones (>5,000 m a.s.l.). Of these, over 700 species have documented medicinal uses in traditional Ayurvedic and Tibetan medical systems. Several Nepali medicinal plants have been ethnopharmacologically associated with anticancer activity and are candidates for systematic computational evaluation as potential EGFR inhibitor sources.

Seven species were selected for this study on the basis of documented biological activity, accessibility in the COCONUT 2.0 database, and relevance to respiratory and oncological conditions:

- ***Swertia chirayita* (Chirayito)**: A bitter alpine herb extensively used in Ayurvedic medicine for fever and liver conditions. Contains xanthenes and secoiridoids with documented kinase-modulatory activity.
- ***Rhododendron arboreum* (Lali Gurans)**: The national flower of Nepal; rhododendron species contain pentacyclic triterpenoids and flavonoids with anti-inflammatory and cytotoxic properties.
- ***Nardostachys jatamansi* (Jatamansi)**: A Himalayan valerian species yielding nardosinone and patchouli-alcohol-type sesquiterpenes; used for neurological and cardiovascular conditions.
- ***Terminalia chebula* (Haritaki)**: One of the three constituents of the classical Ayurvedic formulation Triphala; contains hydrolysable tannins (chebulinic acid, chebulagic acid) with antioxidant and antiproliferative activity.
- ***Berberis aristata* (Chutro)**: Source of berberine, a protoberberine isoquinoline alkaloid with established anticancer activity and documented EGFR pathway modulation.
- ***Ocimum sanctum* (Tulsi)**: The sacred basil of the Indian subcontinent; contains eugenol, ursolic acid, and rosmarinic acid, with documented anticancer and EGFR-inhibitory properties.
- ***Tinospora cordifolia* (Giloy/Gurjo)**: A climbing shrub yielding tinosporin, palmatine, and berberine; used for immunomodulation and anti-inflammatory applications with documented cytotoxic activity.

## 2.6.2 The COCONUT 2.0 Database

The **CO**llection of **O**pen **N**atural **P**rodUcTs (COCONUT) is the largest freely available database of natural product structures, curated and maintained by the Steinbeck group at Friedrich Schiller University Jena (Sorokina et al., 2021). COCONUT 2.0 contains over 400,000 unique natural product structures drawn from more than 50 public natural product databases, each associated with organism taxonomy, literature references, and precomputed molecular descriptors. Structures are provided as canonical SMILES and InChI strings with CAS registry numbers where available.

For this study, COCONUT 2.0 was used to retrieve SMILES for compounds annotated to the seven Nepali plant species of interest using organism taxonomy matching. Retrieved compounds were then subjected to the same preprocessing pipeline as the ChEMBL training data (salt removal, SMILES canonicalisation,  $MW \leq 1000$  Da, RDKit validation) to ensure featurisation consistency.

## 2.7 Virtual Screening in Drug Discovery

### 2.7.1 Overview of Virtual Screening

Virtual screening (VS) is a computational strategy employed at the early stage of drug discovery to prioritise a large library of candidate compounds for experimental testing, based on predicted interaction with a biological target or predicted biological activity. VS substantially reduces the number of compounds that need to be synthesised and assayed experimentally, focusing experimental resources on the most promising candidates (Walters & Murcko, 2020).

Two principal paradigms of VS are distinguished:

- **Structure-based virtual screening (SBVS):** Requires a three-dimensional crystal or cryo-EM structure of the target protein. Molecular docking algorithms (AutoDock Vina, Glide, GOLD) predict the binding pose and estimate the binding affinity of each candidate compound within the target’s binding site. SBVS is accurate but computationally intensive and requires high-resolution structural data.
- **Ligand-based virtual screening (LBVS):** Does not require target structural knowledge. A model trained on known active and inactive compounds (e.g., from ChEMBL) is used to score and rank external compound libraries by predicted activity probability. LBVS is computationally scalable to millions of compounds and is particularly powerful when high-quality bioactivity training data is available.

This thesis implements a **ligand-based virtual screening** pipeline using its two best-performing trained classifiers—the best graph neural network (GCN) and a Random Forest baseline—as the scoring functions, and prioritising leads by cross-model consensus rather than relying on any single model.

### 2.7.2 GNN-Based Virtual Screening

The landmark application of GNN-based virtual screening in drug discovery was the discovery of **halicin** by Stokes et al. (Stokes et al., 2020) at the Collins Laboratory (MIT). A directed-message passing neural network (DMPNN) was trained on 2,335 compounds from the Drug Repurposing Hub to predict broad-spectrum antibiotic activity. The trained model was applied to score over 100 million compounds from the ZINC15 database; halicin (SU-3327, a c-Jun N-terminal kinase inhibitor) was identified as a top-ranked hit and subsequently validated as a potent, broad-spectrum antibiotic in mouse models of *Acinetobacter baumannii* infection. This study demonstrated that GNN-based LBVS can identify structurally novel bioactive compounds that escape detection by conventional similarity searches.

In the kinase inhibitor domain, GNN-based VS has been applied to several targets including EGFR, CDK2, and VEGFR2, typically achieving hit rates of 10–50% in subsequent experimental validation—substantially higher than the 0.1–1% hit rates typical of unguided high-throughput screening (Sun et al., 2020).

### 2.7.3 Natural Product Libraries and Nepali Medicinal Plants

A key methodological choice in virtual screening is the source and composition of the screening library. This thesis employs a **natural product library** strategy: bioactive compounds from seven traditional Nepali medicinal plant species are retrieved from the COCONUT 2.0 database and used as the screening library. The biological and ethnopharmacological rationale is well-established: plants such as *Swertia chirayita*, *Berberis aristata*, and *Ocimum sanctum* have centuries of documented use in the treatment of cancer-related symptoms and have yielded structurally diverse secondary metabolites—alkaloids, flavonoids, terpenes, and tannins—that represent structurally distinct chemotypes compared to synthetic kinase inhibitors, potentially offering novel scaffolds capable of engaging the EGFR ATP-binding pocket through unconventional binding modes. This strategy directly connects computational drug discovery to the biodiversity resources and traditional knowledge of Nepal, supporting affordable and regionally accessible cancer therapeutic research.

### 2.7.4 ADMET Properties and Lead Prioritisation

Virtual screening hit lists require secondary filtering by predicted ADMET (Absorption, Distribution, Metabolism, Excretion, Toxicity) properties to prioritise compounds with the greatest potential for clinical development (Walters & Murcko, 2020). In this study, ADMET-proxy properties were computed using RDKit as physicochemical surrogates:

- **Lipinski’s Rule of Five (Ro5):**  $MW \leq 500$  Da,  $\text{LogP} \leq 5$ ,  $\text{HBD} \leq 5$ ,  $\text{HBA} \leq 10$  as necessary (not sufficient) conditions for oral bioavailability (Lipinski, 2001).
- **Quantitative Estimate of Drug-likeness (QED):** A composite drug-likeness score (range 0–1) integrating MW, LogP, HBD, HBA, TPSA, number of aromatic rings, number of rotatable bonds, and the presence of structural alerts, calibrated to reflect the desirability profile of approved oral drugs (Bickerton et al., 2012).

- **Topological Polar Surface Area (TPSA):** A molecular surface area descriptor correlated with oral absorption and blood-brain barrier permeability. Values  $\leq 140 \text{ \AA}^2$  are associated with adequate gastrointestinal absorption.

## 2.8 Research Gap

A systematic analysis of the reviewed literature reveals three persistent and interrelated methodological gaps in the application of Graph Neural Networks to EGFR inhibitor discovery. These gaps collectively limit the clinical utility and scientific rigour of existing computational approaches:

1. **Evaluation Bias from Random Data Splitting.** The majority of published GNN models for EGFR or related kinase bioactivity prediction employ random train/test splitting (Jiang et al., 2021; Malik et al., 2025). This practice allows molecules with near-identical Bemis-Murcko scaffolds to appear simultaneously in training and test partitions, creating an optimistic bias in reported performance metrics. A model that achieves 0.90 AUROC under random splitting may perform significantly worse when deployed on a genuinely novel scaffold—precisely the scenario encountered in prospective virtual screening. Scaffold-based splitting, which groups all molecules sharing the same core framework into a single partition, provides a more realistic and demanding assessment of model generalisation (Wu et al., 2018).
2. **Incomplete Molecular Featurisation.** Published GNN models for EGFR inhibitor classification frequently employ minimal node feature sets (e.g., atom type only, or a small number of atomic properties) and rarely incorporate explicit, multi-dimensional bond-level (edge) feature vectors (Malik et al., 2025). Bond features encoding stereochemistry, conjugation, and aromaticity carry significant pharmacological information—for example, the aromatic bonds of the quinazoline hinge-binder scaffold and the acrylamide double bond of covalent inhibitors—that is discarded or only coarsely approximated by node-only models or ECFP fingerprints.
3. **Lack of Prospective Screening on Natural Product and Regionally Relevant Compound Libraries.** Most academic GNN papers in the EGFR and kinase inhibitor domain report performance metrics on held-out test sets but do not deploy the trained model in a prospective virtual screening pipeline applied to natural product or ethnopharmacological compound collections (Malik et al., 2025; Sun et al., 2020). For South Asian research communities where EGFR-mutant NSCLC is prevalent but novel targeted therapies are expensive and inaccessible, virtual screening of traditional medicinal plant compounds represents a high-impact translational application that is entirely absent from the computational EGFR inhibitor literature.

Table 1 formally positions this thesis against representative published GNN studies for EGFR inhibitor prediction.

Table 1: Comparison of this thesis against representative published GNN-based EGFR inhibitor classification studies across three methodological dimensions. ✓ = fully addressed; ◦ = partially addressed; × = not addressed.

| Study                            | Scaffold Split | Rich Features | Natural Product Screening |
|----------------------------------|----------------|---------------|---------------------------|
| DeepEGFR (Malik et al., 2025)    | ×              | ◦             | ×                         |
| GCN QSAR (Jiang et al., 2021)    | ×              | ◦             | ×                         |
| AttentiveFP (Xiong et al., 2020) | ×              | ✓             | ×                         |
| Hu et al. (Hu et al., 2020)      | ◦              | ✓             | ×                         |
| <b>This Thesis</b>               | ✓              | ✓             | ✓                         |

This thesis directly and simultaneously addresses all three gaps through a single, cohesive, reproducible open-source computational pipeline combining: (i) Bemis-Murcko scaffold-aware data splitting; (ii) comprehensive 29-dimensional node and 12-dimensional edge featurisation; (iii) Bayesian hyperparameter optimisation via Optuna; and (iv) prospective virtual screening of Nepali medicinal plant compounds from COCONUT 2.0 with ADMET-proxy prioritisation.

## Chapter 3

### Materials and Methods

#### 3.1 System Flow Diagram of the Computational Framework

##### 3.1.1 Overview of the Computational Framework

The overall methodology follows a systematic five-phase pipeline as illustrated in Figure 3. All code was implemented in Python 3.14 using exclusively open-source libraries and executed on a macOS system (CPU).

#### 3.2 Data Acquisition

##### 3.2.1 Target Selection

The primary target for bioactivity data retrieval was the Epidermal Growth Factor Receptor (EGFR), indexed in ChEMBL as target identifier ChEMBL203. EGFR was selected as the model target because it is a clinically validated oncology target in NSCLC, and ChEMBL203 aggregates the largest publicly available collection of quantitative EGFR inhibition bioactivity measurements.

##### 3.2.2 Bioactivity Data Retrieval

Bioactivity data was retrieved programmatically from the ChEMBL database using the `chembl-webresource-client` Python library, which queries the then-current ChEMBL release through the REST API (retrieved May 2026). The query was restricted to compounds satisfying the following criteria:

- **Standard type:** IC<sub>50</sub> (half-maximal inhibitory concentration)
- **Standard relation:** Exact value only (=)

Records with missing SMILES strings, missing IC<sub>50</sub> values, or non-exact relations ( $>$ ,  $<$ ,  $\geq$ ,  $\leq$ ) were excluded. IC<sub>50</sub> values reported in ChEMBL for this target are in nanomolar (nM) units, as assumed throughout the pIC<sub>50</sub> conversion. The raw dataset contained approximately 19,000 compound-activity pairs prior to deduplication and cleaning.

##### 3.2.3 Data Cleaning and Labelling

The raw IC<sub>50</sub> values (in nanomolar units) were converted to the logarithmic pIC<sub>50</sub> scale by taking the negative base-ten logarithm of the molar concentration, so that more potent compounds receive higher pIC<sub>50</sub> scores. Binary activity labels were then assigned using the standard medicinal chemistry threshold: compounds with a pIC<sub>50</sub> of at least 6.0 were labelled *active*, and those below 6.0 were labelled *inactive*. This threshold corresponds

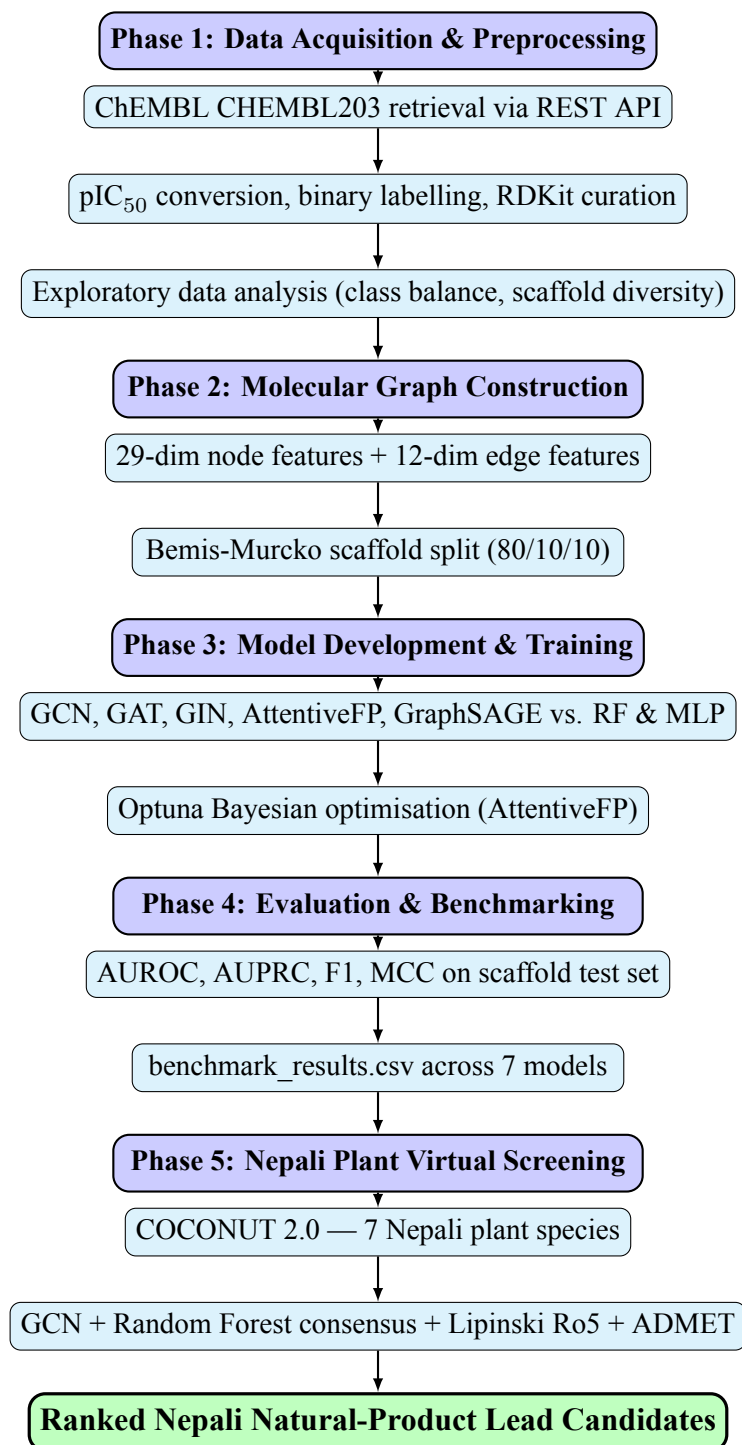


Figure 3: Five-phase computational pipeline for GNN-based EGFR inhibitor classification and Nepali medicinal plant virtual screening.

to an  $IC_{50}$  of  $\leq 1 \mu M$ . For compounds with multiple reported measurements, the median  $pIC_{50}$  was used to obtain a single robust value per molecule.

### 3.2.4 Molecular Standardisation and Filtering

Molecular standardisation was performed using RDKit (version 2026.3.2):

- **Salt removal:** The largest fragment was retained for each compound.
- **Canonicalisation:** SMILES strings were canonicalised to ensure unique molecular representation.
- **Molecular weight filter:** Compounds with  $MW > 1000$  Da were excluded.
- **Deduplication:** Duplicate canonical SMILES were collapsed using the median  $pIC_{50}$ .

The final curated dataset comprised **10,523 unique compounds**, distributed as 7,792 actives (74.0%) and 2,731 inactives (26.0%).

### 3.2.5 Exploratory Data Analysis

Prior to molecular graph construction, an exploratory data analysis (EDA) step was performed on the cleaned dataset to characterise its statistical structure and inform modelling decisions. Using RDKit and pandas, the analysis quantified the class balance, the distribution of  $pIC_{50}$  potency values, the ranges of the physicochemical properties (molecular weight, LogP), and the Bemis-Murcko scaffold diversity (number of unique scaffolds and the proportion of singleton scaffolds). This characterisation, reported in Chapter 4, both confirms the integrity of the curated dataset and motivates the use of scaffold-based splitting, given the long-tailed scaffold distribution observed.

## 3.3 Molecular Graph Construction

### 3.3.1 Graph Representation

Each molecule was converted to an undirected graph using PyTorch Geometric (version 2.7.0), where every node represents an atom and every edge represents a chemical bond (the molecule-to-graph conversion is illustrated in Figure 1). Each atom node carries a 29-dimensional feature vector and each bond edge a 12-dimensional feature vector, detailed below.

#### 3.3.2 Node Features

Each atom was represented by a 29-dimensional feature vector encoding:

- **Atom type** (one-hot, 10 dim): C, N, O, S, F, Cl, Br, I, P, Unknown
- **Atom degree** (one-hot, 6 dim): 0, 1, 2, 3, 4,  $\geq 5$
- **Formal charge** (1 dim): integer value

- **Hydrogen count** (one-hot, 5 dim): 0, 1, 2, 3,  $\geq 4$
- **Hybridisation** (one-hot, 5 dim): SP, SP2, SP3, SP3D, SP3D2
- **Aromaticity** (binary, 1 dim)
- **Ring membership** (binary, 1 dim)

### 3.3.3 Edge Features

Each bond was represented by a 12-dimensional edge feature vector encoding:

- **Bond type** (one-hot, 4 dim): Single, Double, Triple, Aromatic
- **Conjugation** (binary, 1 dim)
- **Ring membership** (binary, 1 dim)
- **Stereochemistry** (one-hot, 6 dim): STEREOANY, STEREOZ, STEREOE, STEREOCIS, STEREOTRANS

## 3.4 Data Splitting

To ensure rigorous evaluation on genuinely novel chemical scaffolds, a deterministic **Bemis-Murcko Scaffold Split** was implemented using RDKit. The Bemis-Murcko algorithm extracts the core macrocyclic framework of each molecule by retaining ring systems, linker atoms, and chain atoms connected to rings while removing side chains. Compounds sharing identical scaffold hashes were allocated exclusively to one split partition. The dataset was partitioned as follows:

Table 2: Dataset partitioning using the Bemis-Murcko scaffold split.

| Partition  | Proportion | Compounds | Unique Scaffolds |
|------------|------------|-----------|------------------|
| Training   | 80%        | 8,418     | majority         |
| Validation | 10%        | 1,052     | held-out         |
| Test       | 10%        | 1,053     | held-out         |

## 3.5 Machine Learning Model Development

### 3.5.1 Traditional Baseline Models

Two traditional machine learning baselines were implemented using scikit-learn (version 1.8.0) trained on 2048-bit ECFP4 (Morgan) fingerprints (radius = 2) computed by RDKit:

- **Random Forest (RF)**: An ensemble of 500 decision trees with `class_weight='balanced'` to account for the 74:26 class imbalance. Hyperparameters included `max_features='sqrt'` and `min_samples_leaf=1`.

- **Multi-Layer Perceptron (MLP):** A three-layer neural network ( $2048 \rightarrow 512 \rightarrow 128 \rightarrow 1$ ) with BatchNorm and Dropout(0.3) at each hidden layer, trained on ECFP4 fingerprints with BCEWithLogitsLoss (pos\_weight), Adam optimiser, and early stopping (patience = 10) on an internal 90/10 validation split. This evaluates whether deep learning on fingerprint features alone offers benefits beyond the Random Forest ensemble.

### 3.5.2 Graph Neural Network Architectures

Five GNN architectures were implemented using PyTorch Geometric:

#### 3.5.2.1 Graph Convolutional Network (GCN)

The GCN propagates information using degree-normalised spectral convolution. At each layer, every atom embedding is updated by averaging the transformed embeddings of its bonded neighbours and itself, where the averaging weights are normalised by the degrees of the participating atoms (with added self-loops). This symmetric normalisation prevents high-degree atoms from dominating the representation. Three convolutional layers were stacked, each followed by batch normalisation, a ReLU non-linearity, and dropout.

#### 3.5.2.2 Graph Attention Network (GAT)

The GAT replaces the fixed normalisation of the GCN with learnable attention weights: for each atom, the contribution of every neighbour is weighted by an attention coefficient computed from the two atoms' embeddings and the connecting bond features, then normalised across all neighbours. This lets the network learn which bonds and substructures are most relevant to activity. Four attention heads were used in the intermediate layers (their outputs concatenated) and a single averaged head in the final layer. Bond (edge) features were supplied directly to the attention mechanism through the GATv2 formulation.

#### 3.5.2.3 Graph Isomorphism Network (GIN)

GIN is the most theoretically expressive message-passing scheme within the Weisfeiler–Lehman framework. Rather than averaging, each atom embedding is updated by summing its neighbours' embeddings (preserving multiset information that mean or max aggregation would discard), adding the atom's own embedding, and passing the result through a two-layer multilayer perceptron. The GINEConv variant used here extends this update to incorporate the 12-dimensional bond feature vectors.

#### 3.5.2.4 GraphSAGE

GraphSAGE (Hamilton et al., 2017) updates each atom by combining its own previous embedding with the mean of its neighbours' embeddings and passing the combination through a learnable transformation. Because the update depends only on local neighbourhood aggregation rather than the global graph structure, GraphSAGE is inherently inductive and generalises naturally to unseen molecules. Three SAGEConv layers with batch normalisation and ReLU activation are used, followed by concatenated global

mean and max pooling and a linear classification head. GraphSAGE was included to provide an inductive neighbourhood-sampling baseline that is evaluated under the same scaffold split conditions as the attention-based models.

#### 3.5.2.5 AttentiveFP

AttentiveFP implements a two-level attention mechanism. At the atom level, a graph attention layer aggregates neighbourhood information weighted by learned atom-atom attention scores. At the molecule level, a Gated Recurrent Unit (GRU)-based readout iteratively refines a single global molecular embedding over several timesteps, allowing the model to focus on the substructures most predictive of activity. Crucially, AttentiveFP natively accepts both the atom and bond feature vectors without preprocessing, making it uniquely suited to exploit the full 29-dimensional node and 12-dimensional edge feature specification.

#### 3.5.3 Training Protocol

All GNN models were trained with the following protocol:

- **Loss function:** Binary cross-entropy with logits incorporating a positive class weight (the ratio of negative to positive training examples, approximately 0.34) to address the class imbalance.
- **Optimiser:** Adam with its default momentum parameters.
- **Scheduler:** ReduceLROnPlateau with reduction factor 0.5 and patience 10 epochs.
- **Early stopping:** Training halted if validation AUROC did not improve for 20 consecutive epochs (30 for the tuned AttentiveFP).
- **Batch size:** 64 for default training; selected by Optuna for the tuned AttentiveFP.
- **Readout:** Global mean and max pooling were concatenated to form the graph-level representation.

#### 3.5.4 Model Evaluation Metrics

Models were evaluated using the following metrics on the scaffold-held-out test set:

- **AUROC:** Threshold-independent measure of discriminative ability across the full range of classification thresholds.
- **AUPRC:** Area under the precision-recall curve, more informative than AUROC under class imbalance.
- **F1-Score:** Harmonic mean of precision and recall at the 0.5 decision threshold.
- **Matthews Correlation Coefficient (MCC):** A balanced correlation between predicted and true labels derived from all four confusion-matrix counts (true/false

positives and negatives), ranging from  $-1$  (total disagreement) through  $0$  (random) to  $+1$  (perfect prediction). It is robust to class imbalance and is reported as the primary balanced metric.

- **Accuracy:** Proportion of correctly classified compounds.

### 3.5.5 Hyperparameter Tuning

Bayesian hyperparameter optimisation was performed for the AttentiveFP model using the Optuna framework (version 4.8.0) with the Tree-structured Parzen Estimator (TPE) sampler. The search space encompassed:

Table 3: Hyperparameter search space for AttentiveFP Optuna optimisation.

| Hyperparameter  | Type        | Range / Options       |
|-----------------|-------------|-----------------------|
| hidden_channels | Categorical | {64, 128, 256}        |
| num_layers      | Integer     | 2–4                   |
| num_timesteps   | Integer     | 2–4                   |
| dropout         | Float (log) | 0.1–0.5               |
| batch_size      | Categorical | {32, 64, 128}         |
| learning_rate   | Float (log) | $10^{-4}$ – $10^{-2}$ |
| weight_decay    | Float (log) | $10^{-5}$ – $10^{-3}$ |

The objective function to maximise was validation AUROC. Pruning of unpromising trials was performed using the Optuna MedianPruner.

## 3.6 Nepali Medicinal Plant Virtual Screening

### 3.6.1 Plant Species and Compound Library Curation

A virtual screening library was assembled from the COCONUT 2.0 natural product database (Sorokina et al., 2021) by filtering for compounds annotated to seven traditional Nepali medicinal plant species:

Table 4: Nepali medicinal plant species used for virtual screening.

| Species                       | Common Name | Traditional Use         |
|-------------------------------|-------------|-------------------------|
| <i>Swertia chirayita</i>      | Chirayito   | Fever, liver conditions |
| <i>Rhododendron arboreum</i>  | Lali Gurans | Anti-inflammatory       |
| <i>Nardostachys jatamansi</i> | Jatamansi   | Neurological, cardiac   |
| <i>Terminalia chebula</i>     | Haritaki    | Antioxidant, Triphala   |
| <i>Berberis aristata</i>      | Chutro      | Anticancer (berberine)  |
| <i>Ocimum sanctum</i>         | Tulsi       | Anticancer, anti-EGFR   |
| <i>Tinospora cordifolia</i>   | Giloy/Gurjo | Immunomodulatory        |

Retrieved compounds were subjected to the same preprocessing pipeline as the ChEMBL training data: salt removal (largest fragment retained), SMILES canonicalisation, molecular weight filter ( $MW \leq 1000$  Da), removal of invalid SMILES, and

deduplication by canonical SMILES. ADMET-proxy descriptors (MW, LogP, HBD, HBA, TPSA, QED) were computed using RDKit for all retained compounds.

### 3.6.2 Inference and Prioritisation

To avoid basing lead selection on any single model—and in particular to avoid the weakest classifier—the library was scored with the study’s two best-performing models: the best graph neural network (GCN) and the best overall classifier (Random Forest). For the GCN, each compound was featurised using the identical 29-dimensional node and 12-dimensional edge graph specification as the training data and passed through a forward inference to produce  $\hat{p}_i = P(\text{active} \mid G_i)$ . For the Random Forest, each compound was encoded as a 2048-bit ECFP4 fingerprint and scored with the trained ensemble. Predictions from both models were retained so that consensus (agreement between the two best models) could be assessed. The tuned AttentiveFP screen was also computed for comparison, exposing its over-confidence relative to the two best-calibrated models.

Compounds were ranked by predicted activity probability  $\hat{p}_i$  under each model. The top leads were further assessed by:

- **Lipinski Rule of Five compliance:**  $MW \leq 500$  Da,  $\text{LogP} \leq 5$ ,  $\text{HBD} \leq 5$ ,  $\text{HBA} \leq 10$  (necessary condition for oral bioavailability).
- **QED score:** A composite drug-likeness score targeting  $\text{QED} \geq 0.40$ .
- **TPSA:** Values  $\leq 140 \text{ \AA}^2$  for adequate oral absorption.

The final outputs are prioritised ranked lists for each best model (`results/nepali_leads_rf.csv` and `results/nepali_leads_gcn.csv`) of Nepali natural-product-derived compounds with computed ADMET-proxy properties, representing actionable candidates for experimental EGFR inhibition assays.

## Chapter 4

### Results and Discussion

This chapter presents the complete experimental results of the five-phase computational pipeline. Section 4.1 characterises the curated EGFR bioactivity dataset. Section 4.2 evaluates and compares all seven classification models. Section 4.3 reports Nepali medicinal plant virtual screening outcomes and lead prioritisation. Section 4.4 provides an integrated discussion of all findings.

#### 4.1 Data Distribution Analysis

##### 4.1.1 Dataset Composition and Class Balance

The curated dataset comprised **10,523 unique EGFR-bioactive compounds** retrieved from ChEMBL (ChEMBL203) following strict quality filtering. The binary label distribution exhibited a moderate class imbalance: **7,792 active compounds** (74.0%,  $\text{pIC}_{50} \geq 6.0$ ) and **2,731 inactive compounds** (26.0%,  $\text{pIC}_{50} < 6.0$ ). This imbalance is inherent to bioactivity databases and reflects the practical reality that published compounds are disproportionately those that demonstrated measurable activity in assay screening campaigns.

Table 5 summarises the dataset composition across the three scaffold-split partitions.

Table 5: Dataset composition and class distribution across scaffold-split partitions.

| Partition    | Total         | Active       | Inactive     | Active (%)  |
|--------------|---------------|--------------|--------------|-------------|
| Training     | 8,418         | 6,292        | 2,126        | 74.7        |
| Validation   | 1,052         | 743          | 309          | 70.6        |
| Test         | 1,053         | 757          | 296          | 71.9        |
| <b>Total</b> | <b>10,523</b> | <b>7,792</b> | <b>2,731</b> | <b>74.0</b> |

The class balance remained broadly consistent across all three partitions (74.7% / 70.6% / 71.9%), with the held-out validation and test partitions being marginally less active-enriched. This is expected under a strict Bemis-Murcko scaffold split, in which the large shared scaffold families are concentrated in the training set and the held-out partitions are dominated by novel singleton scaffolds. The overall class balance is shown in Figure 4.

##### 4.1.2 $\text{pIC}_{50}$ Distribution

The  $\text{pIC}_{50}$  distribution across the full dataset exhibited a bimodal profile. The active subpopulation was centred around  $\text{pIC}_{50} \approx 7.5\text{--}8.0$  (corresponding to  $\text{IC}_{50}$  values of

EGFR Dataset Class Balance

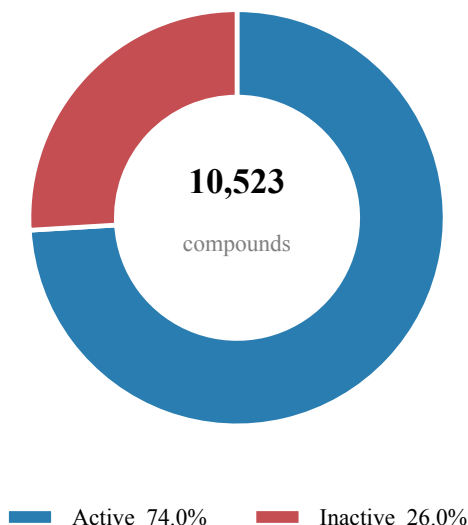


Figure 4: Class balance of the curated EGFR dataset: 74.0% active and 26.0% inactive compounds across 10,523 unique molecules.

10–32 nM), reflecting the high-potency compounds typically selected for further optimisation in medicinal chemistry campaigns. The inactive subpopulation showed a peak near  $\text{pIC}_{50} \approx 5.2\text{--}5.5$ , representing weak binders at the micro-molar level. The binary threshold at  $\text{pIC}_{50} = 6.0$  ( $\text{IC}_{50} = 1\text{ }\mu\text{M}$ ) was placed in a relatively low-density valley between these two modes, providing a chemically meaningful separation rather than an arbitrary cut-point. This is consistent with the standard medicinal chemistry convention that defines a “hit” as a compound with measurable activity at or below  $1\text{ }\mu\text{M}$ . The bimodal structure and the placement of the decision threshold are shown in Figure 5.

#### 4.1.3 Physicochemical Property Analysis

Analysis of the RDKit-computed physicochemical properties of the curated dataset revealed that the majority of compounds occupied drug-like chemical space. Key observations are summarised in Table 6.

Table 6: Summary statistics of physicochemical properties of the curated dataset.

| Property                | Min  | Median | Mean  | Max   |
|-------------------------|------|--------|-------|-------|
| Molecular Weight (Da)   | 65.4 | 478.0  | 475.0 | 991.7 |
| LogP                    | −2.0 | 4.5    | 4.5   | 13.1  |
| H-Bond Donors           | 0    | 2      | 2.2   | 10    |
| H-Bond Acceptors        | 0    | 7      | 6.6   | 17    |
| TPSA ( $\text{\AA}^2$ ) | 0.0  | 96.3   | 95.2  | 273.0 |

The median molecular weight of 478 Da and median LogP of 4.5 sit close to the upper limits of Lipinski’s Rule of Five (Ro5), consistent with the physicochemical characteris-

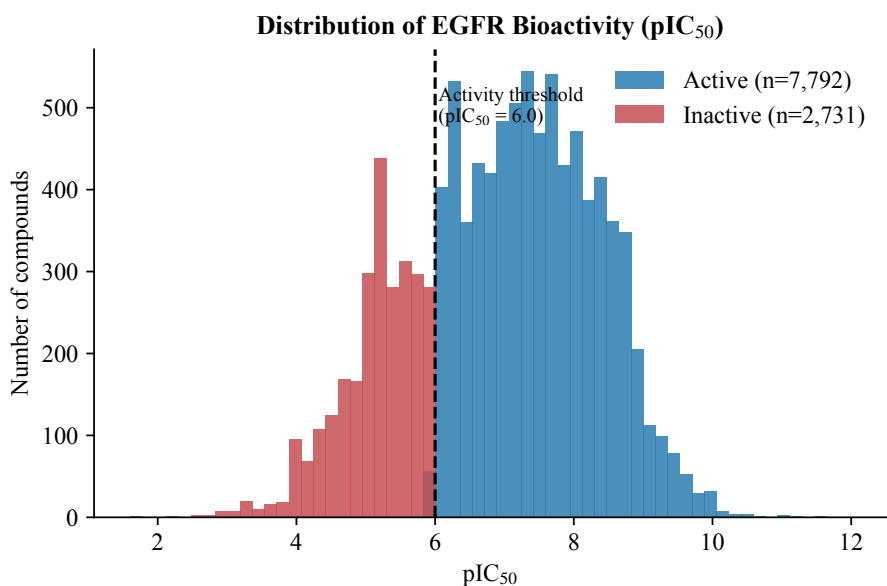


Figure 5: Distribution of EGFR bioactivity ( $\text{pIC}_{50}$ ) for the curated dataset, coloured by activity label. The active and inactive subpopulations form two modes separated by the  $\text{pIC}_{50} = 6.0$  decision threshold (dashed line,  $\text{IC}_{50} = 1 \mu\text{M}$ ).

tics of ATP-competitive kinase inhibitors, which frequently occupy the upper boundary of drug-like chemical space. Approximately **47%** of the compounds were fully Lipinski-compliant ( $\text{MW} \leq 500 \text{ Da}$ ,  $\text{LogP} \leq 5$ ,  $\text{HBD} \leq 5$ ,  $\text{HBA} \leq 10$ ); the remainder most often exceed the  $\text{LogP}$  or molecular weight thresholds, reflecting the lipophilic, higher-mass character typical of this inhibitor class. The molecular weight and lipophilicity distributions are shown in Figure 6.

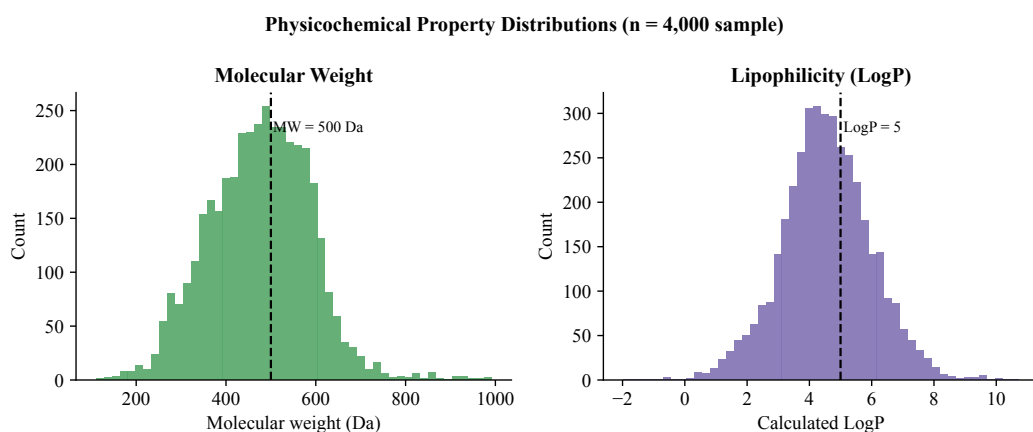


Figure 6: Distributions of molecular weight (left) and calculated  $\text{LogP}$  (right) for a random sample of the curated dataset. Dashed lines mark the Lipinski thresholds ( $\text{MW} = 500 \text{ Da}$ ,  $\text{LogP} = 5$ ); most compounds cluster near these thresholds, characteristic of kinase inhibitors.

#### 4.1.4 Scaffold Diversity Analysis

The Bemis-Murcko scaffold analysis identified **3,693 unique scaffolds** across the full dataset. Of these, 2,418 (65.5%) were singletons, comprising only a single compound. The most frequently occurring scaffold, observed in 572 compounds, corresponds to the 4-anilinoquinazoline core, which is the defining pharmacophore of first- and second-generation EGFR TKIs including gefitinib, erlotinib, and lapatinib. Several further scaffolds each comprising over one hundred compounds include the pyrrolopyrimidine core characteristic of certain second-generation and third-generation inhibitors. This scaffold-level analysis confirms that the dataset represents a structurally diverse yet pharmacophore-focused chemical space. Figure 7 shows the most frequent scaffolds and the overall split between singleton and multi-compound scaffolds, which motivates the use of scaffold-based data splitting.

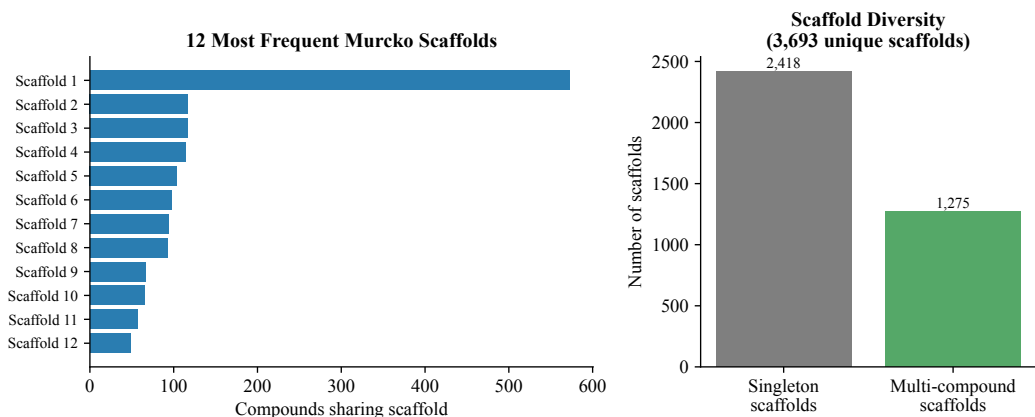


Figure 7: Bemis-Murcko scaffold diversity of the curated dataset. Left: number of compounds sharing each of the most frequent scaffolds. Right: the split between singleton scaffolds (a single compound) and multi-compound scaffolds, illustrating the long-tailed scaffold distribution that necessitates scaffold-based splitting.

## 4.2 Machine Learning Model Performance

### 4.2.1 Individual Model Performance on the Scaffold Test Set

All seven models were trained under identical scaffold split conditions and evaluated on the held-out scaffold test set of 1,053 compounds. Results are presented in Table 7. The classification threshold was set at  $\hat{p} = 0.5$  for all threshold-dependent metrics (F1-Score, MCC, Accuracy).

Table 7: Performance comparison of all models on the scaffold-held-out test set (threshold = 0.50 for F1, MCC, Accuracy).

| Model         | Type       | AUROC  | AUPRC  | F1     | MCC    | Accuracy |
|---------------|------------|--------|--------|--------|--------|----------|
| Random Forest | ML + ECFP4 | 0.8827 | 0.9496 | 0.8762 | 0.5072 | 0.8129   |
| MLP           | ML + ECFP4 | 0.8547 | 0.9354 | 0.8629 | 0.5281 | 0.8053   |
| GCN           | GNN        | 0.8454 | 0.9332 | 0.7877 | 0.4707 | 0.7312   |
| GraphSAGE     | GNN        | 0.8229 | 0.9179 | 0.8070 | 0.4706 | 0.7474   |
| GIN           | GNN        | 0.8217 | 0.9163 | 0.8487 | 0.4865 | 0.7863   |
| GAT           | GNN        | 0.8162 | 0.9176 | 0.7689 | 0.4473 | 0.7123   |
| AttentiveFP   | GNN        | 0.8087 | 0.9108 | 0.8254 | 0.3989 | 0.7521   |

Figure 8 visualises these results across the four headline metrics, making the relative ordering of the seven models directly comparable.

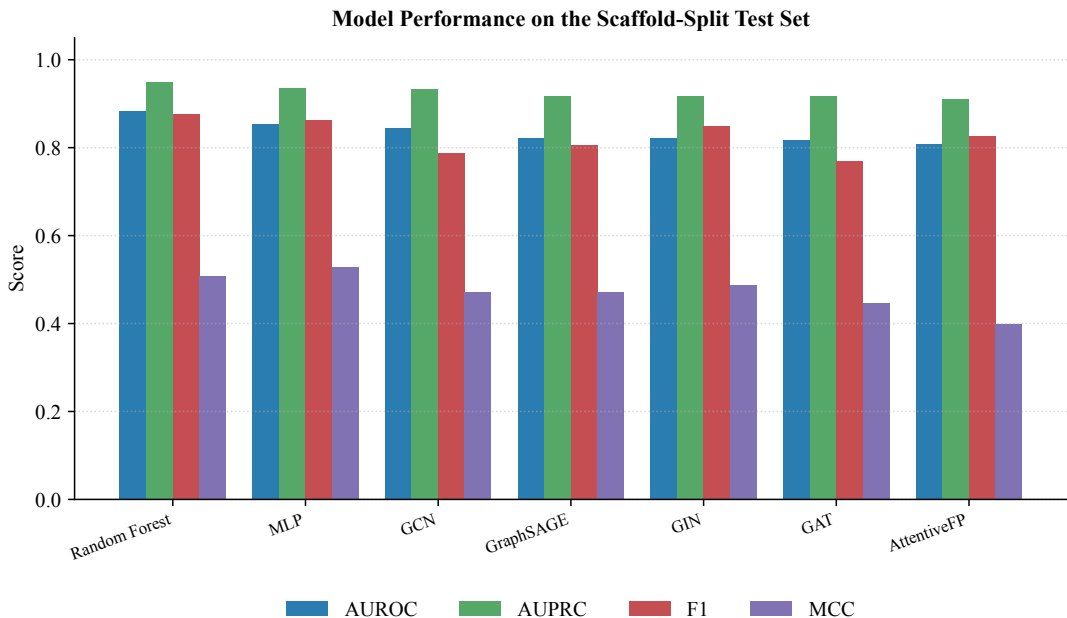


Figure 8: Performance of all seven models on the scaffold-held-out test set across AUROC, AUPRC, F1-score, and MCC. The Random Forest and MLP fingerprint baselines lead on AUROC, while the graph neural networks form a competitive cluster led by GCN.

#### 4.2.1.1 Random Forest Performance

Random Forest achieved the highest test-set AUROC (0.8827) and AUPRC (0.9496) among all models, together with the strongest F1-Score (0.8762); the MLP baseline attained the highest MCC (0.5281), narrowly ahead of Random Forest (0.5072). The strong performance of the two fingerprint-based baselines is consistent with established observations in cheminformatics benchmarks: ensemble and shallow neural methods built on ECFP4 fingerprints are highly competitive on bioactivity datasets of this scale (~10,000 compounds), particularly when the training and test chemical spaces share

fingerprint-level structural similarity. The high AUPRC value (0.9496) is especially significant given the 74:26 class imbalance, indicating that Random Forest maintains high precision at high recall thresholds.

#### 4.2.1.2 Graph Neural Network Performance

Among the five GNN architectures at default settings, **GCN achieved the highest AUROC (0.8454)**, followed by GraphSAGE (0.8229), GIN (0.8217), GAT (0.8162), and AttentiveFP (0.8087). Notably, the simpler convolution- and sampling-based architectures outperformed the more elaborate attention-based models on this metric—a common observation, since high-capacity, molecule-specific architectures such as AttentiveFP are highly sensitive to hyperparameter choices and tend to underperform simpler models until properly tuned (Section 4.2.2). The per-architecture characteristics are:

- **GCN** provided the strongest and most stable default GNN, its degree-normalised convolution generalising well to novel scaffolds without extensive tuning.
- **GraphSAGE** performed comparably (AUROC 0.8229), with its concatenated self-and-neighbour update giving robust inductive generalisation on molecular graphs of this scale.
- **GIN** achieved the strongest F1-Score (0.8487) and MCC (0.4865) among the GNNs; its sum aggregation preserves neighbourhood multiset information that benefits threshold-dependent classification.
- **GAT** did not translate its learned attention weights into a higher AUROC than GCN at default settings, indicating that the additional parameters require tuning to pay off.
- **AttentiveFP**, despite its sophisticated two-level attention and native 12-dimensional edge-feature integration, recorded the lowest default AUROC (0.8087) and MCC (0.3989). This points to substantial untapped capacity that motivates the dedicated hyperparameter optimisation reported in Section 4.2.2.

#### 4.2.1.3 GNN vs. Traditional ML Baselines

The performance gap between the fingerprint baselines and all default GNN models (AUROC difference: 0.037–0.074 relative to Random Forest) reflects a well-documented phenomenon in molecular machine learning: GNNs require larger training datasets and careful tuning to overcome the strong inductive bias provided by ECFP4 fingerprints. Notably, the MLP baseline (AUROC 0.8547), which applies deep learning to the same ECFP4 fingerprints, outperformed all five default GNN models, confirming that the fixed fingerprint representation—rather than deep learning capacity alone—drives the advantage of the baselines. The ECFP4 representation was refined through decades of cheminformatics research specifically for molecular activity prediction, while GNN architectures must learn to identify relevant substructures from first principles. This gap is not closed by hyperparameter optimisation: tuning AttentiveFP (Section 4.2.2) improved its validation AUROC but left its test AUROC essentially unchanged, so on the strict scaffold test set the fingerprint baselines remained ahead of every graph neural network.

### 4.2.2 Hyperparameter Tuning Results

Bayesian hyperparameter optimisation was performed over 20 Optuna trials using the Tree-structured Parzen Estimator sampler, with the objective of maximising validation AUROC. The optimal hyperparameter configuration identified for AttentiveFP is presented in Table 8.

Table 8: Optimal AttentiveFP hyperparameters identified by Optuna Bayesian optimisation over 20 trials, with the resulting performance.

| Hyperparameter                               | Default Value | Optimal Value         |
|----------------------------------------------|---------------|-----------------------|
| hidden_channels                              | 128           | 128                   |
| num_layers                                   | 2             | 3                     |
| num_timesteps                                | 2             | 2                     |
| dropout                                      | 0.30          | 0.129                 |
| batch_size                                   | 64            | 64                    |
| learning_rate                                | 0.001         | $2.61 \times 10^{-3}$ |
| weight_decay                                 | 0.0           | $1.10 \times 10^{-5}$ |
| <b>Best validation AUROC (20 trials)</b>     |               | <b>0.8632</b>         |
| <b>Tuned test AUROC (cf. default 0.8087)</b> |               | <b>0.7970</b>         |

The optimisation raised the best *validation* AUROC to **0.8632**, a modest improvement that lifts AttentiveFP into the middle of the field on the validation metric. On the held-out scaffold *test* set, however, the tuned model achieved an AUROC of only **0.7970**—essentially unchanged from the untuned model (0.8087), still the lowest of the GNNs, and well below the Random Forest (0.8827) and MLP (0.8547) fingerprint baselines. In other words, hyperparameter tuning improves the validation-set fit but does not translate into better test-set generalisation, and the gap to the fingerprint-based baselines is not closed on this strict scaffold-split task—a realistic outcome consistent with the well-documented difficulty of out-performing tuned fingerprint models on medium-sized bioactivity datasets.

The selected hyperparameters are nonetheless informative:

- **num\_layers = 3**: A third message-passing layer enables atoms to exchange information across up to three bonds, covering the spatial extent of most pharmacophoric motifs (the default used only two).
- **dropout = 0.129**: A lower dropout than the default (0.30) was preferred, indicating that the model benefits from retaining more of its representational capacity on this dataset.
- **learning\_rate =  $2.61 \times 10^{-3}$** : A higher learning rate than the default ( $10^{-3}$ ) accelerated convergence within the per-trial epoch budget.
- **weight\_decay =  $1.10 \times 10^{-5}$** : A small non-zero L2 penalty (absent by default) provided mild regularisation against scaffold-specific overfitting.

- **hidden\_channels**, **num\_timesteps**, and **batch\_size** were retained at their default values (128, 2, and 64), indicating these were already near-optimal for the task.

### 4.3 Nepali Medicinal Plant Virtual Screening Results

#### 4.3.1 Screening Library Composition

The Nepali medicinal plant screening library was assembled from the COCONUT 2.0 database by filtering for compounds annotated to seven traditional Nepali plant species, then applying the same RDKit standardisation pipeline as the ChEMBL training data. After salt removal, canonicalisation, the molecular-weight filter, and deduplication by canonical SMILES, the final library comprised **535 unique natural product compounds** distributed across the seven species as summarised in Table 9.

Table 9: Composition of the Nepali medicinal plant screening library from COCONUT 2.0.

| Plant Species                 | Compound Class               | Compounds  |
|-------------------------------|------------------------------|------------|
| <i>Nardostachys jatamansi</i> | Sesquiterpenes, coumarins    | 203        |
| <i>Terminalia chebula</i>     | Hydrolysable tannins         | 195        |
| <i>Tinospora cordifolia</i>   | Diterpenes, alkaloids        | 71         |
| <i>Swertia chirayita</i>      | Xanthenes, secoiridoids      | 23         |
| <i>Berberis aristata</i>      | Isoquinoline alkaloids       | 19         |
| <i>Rhododendron arboreum</i>  | Triterpenoids, flavonoids    | 15         |
| <i>Ocimum sanctum</i>         | Phenylpropanoids, terpenoids | 9          |
| <b>Total</b>                  | —                            | <b>535</b> |

The library is dominated by *Nardostachys jatamansi* (203 compounds) and *Terminalia chebula* (195), reflecting their extensive phytochemical characterisation in COCONUT, whereas *Ocimum sanctum* (9) and *Rhododendron arboreum* (15) are comparatively under-represented—an illustration of the uneven database coverage of Nepali medicinal flora. The distribution across the seven species is shown in Figure 9.

#### 4.3.2 Multi-Model Screening Funnel and Hit Identification

A preliminary screen scored the library with the tuned AttentiveFP model. However, AttentiveFP was in fact the *weakest* classifier on the scaffold test set (Section 4.2). To avoid basing lead selection on the weakest model, the prospective screen was re-run and deployed with the study’s two *best-performing* models—the strongest graph neural network (GCN, test AUROC 0.8454) and the strongest overall classifier (Random Forest, test AUROC 0.8827)—applied to the identical 535-compound library. Table 10 compares the progressive screening funnel obtained under all three models.

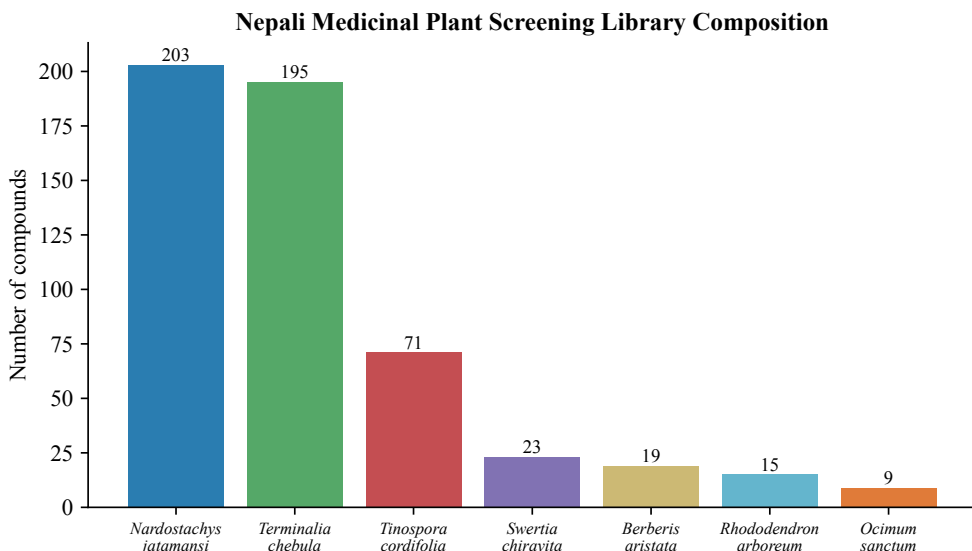


Figure 9: Composition of the Nepali medicinal plant screening library by species, totalling 535 unique natural product compounds retrieved from COCONUT 2.0.

Table 10: Nepali plant virtual screening funnel under the three models, showing the number of compounds retained at each activity and drug-likeness stage. The three models disagree sharply, revealing the model-sensitivity of the screen.

| Filter Stage                                                   | AttentiveFP<br>(0.81) | GCN<br>(0.85) | Random Forest<br>(0.88) |
|----------------------------------------------------------------|-----------------------|---------------|-------------------------|
| Full plant compound library                                    | 535                   | 535           | 535                     |
| Predicted active ( $\hat{p} \geq 0.50$ )                       | 144                   | 24            | 181                     |
| Predicted active ( $\hat{p} \geq 0.80$ )                       | 83                    | 1             | 6                       |
| Predicted active ( $\hat{p} \geq 0.90$ )                       | 71                    | 0             | 1                       |
| Drug-like leads ( $\hat{p} \geq 0.90$ , Ro5, QED $\geq 0.40$ ) | 11                    | 0             | 1                       |

The three funnels differ dramatically, and this divergence is itself a central result. The tuned AttentiveFP predicts 144 actives at  $\hat{p} \geq 0.50$  and an implausible 71 compounds at  $\hat{p} \geq 0.90$ ; the best graph network (GCN) predicts only 24 actives and *not a single* compound at  $\hat{p} \geq 0.90$ ; and the best overall classifier (Random Forest) predicts 181 actives but only one compound at  $\hat{p} \geq 0.90$ . The abundance of apparently high-confidence hits in the AttentiveFP screen is therefore an artefact of poor probability calibration on out-of-distribution natural products by the weakest model, and is *not* reproduced by either best-calibrated model. This model-sensitivity is the principal caution of the screening campaign: virtual-screening hit lists on natural-product chemical space depend strongly on the choice of classifier, and the most trustworthy leads are those nominated by the strongest models—ideally by more than one. A second, related artefact—over-confidence on large, non-drug-like natural products—is also model-dependent in its severity: the weakest model (AttentiveFP) scored large hydrolysable *Terminalia chebula* tannins (e.g., pentagalloylglucose, MW  $\approx$  941 Da) at  $\hat{p} \approx 1.0$ , and the best GNN (GCN) assigned its highest scores to large *Swertia* and *Tinospora* triter-

penoids, all of which violate the Rule of Five. Only the best classifier (Random Forest) placed drug-like compounds—the flavones apigenin and kaempferol—at the very top of its ranking, relegating the tannins below them ( $\hat{p} \approx 0.82$ ). In every case, non-drug-like large compounds still appear among the high scorers, reinforcing the need to couple activity prediction with drug-likeness filtering. The three funnels are compared in Figure 10.

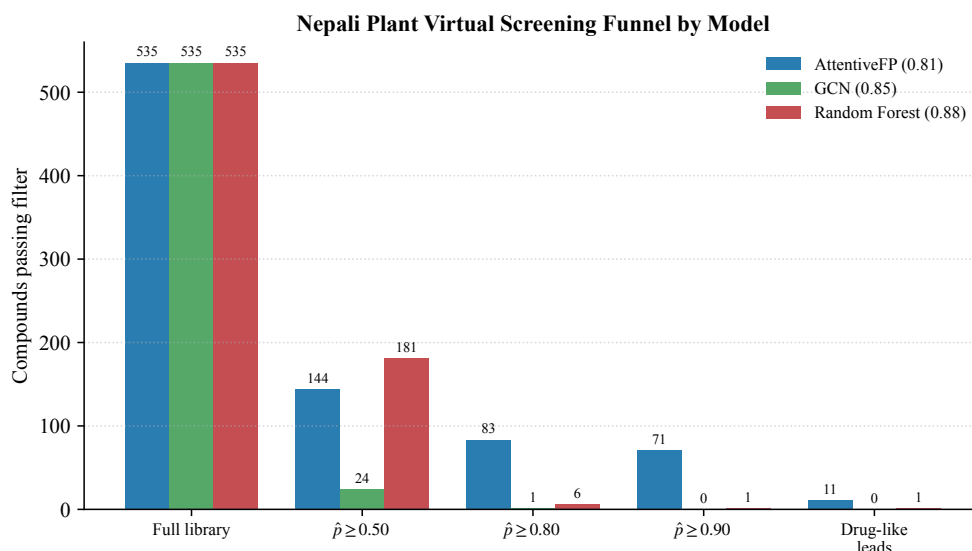


Figure 10: Virtual screening funnel for the Nepali medicinal plant library under the three models (tuned AttentiveFP, best GNN = GCN, best classifier = Random Forest), showing the number of compounds retained at each prioritisation stage. The sharp disagreement between models—AttentiveFP retaining 71 compounds at  $\hat{p} \geq 0.90$  versus 0 for GCN and 1 for Random Forest—illustrates the model-sensitivity of the screen.

### 4.3.3 Consensus Leads from the Best-Performing Models

Because no single classifier is authoritative, lead prioritisation is based on the two best-performing models rather than the weak AttentiveFP screen. The top drug-like (Lipinski-compliant) leads nominated by the Random Forest are listed in Table 11, and those nominated by the GCN in Table 12.

Table 11: Top drug-like (Lipinski-compliant) Nepali plant leads nominated by the best overall classifier (Random Forest), ranked by predicted activity. Listed compounds are those bearing common (trivial) names.

| Rank | Compound       | $\hat{p}$ | MW (Da) | LogP | QED   | Ro5 | Plant                |
|------|----------------|-----------|---------|------|-------|-----|----------------------|
| 1    | Apigenin       | 0.9127    | 270.2   | 2.58 | 0.632 | ✓   | <i>O. sanctum</i>    |
| 2    | Kaempferol     | 0.8340    | 286.2   | 2.28 | 0.546 | ✓   | <i>N. jatamansi</i>  |
| 3    | Isogentisin    | 0.7282    | 258.2   | 2.37 | 0.655 | ✓   | <i>S. chirayita</i>  |
| 4    | Luteolin       | 0.6789    | 286.2   | 2.28 | 0.511 | ✓   | <i>T. chebula</i>    |
| 5    | Tinosporaside  | 0.6607    | 492.5   | 0.63 | 0.440 | ✓   | <i>T. cordifolia</i> |
| 6    | Columbin       | 0.6180    | 358.4   | 2.53 | 0.613 | ✓   | <i>T. cordifolia</i> |
| 7    | Palmarin       | 0.6042    | 374.4   | 1.74 | 0.591 | ✓   | <i>T. cordifolia</i> |
| 8    | Nardostachysin | 0.5836    | 430.5   | 2.95 | 0.526 | ✓   | <i>N. jatamansi</i>  |

Table 12: Top drug-like (Lipinski-compliant) Nepali plant leads nominated by the best graph neural network (GCN), ranked by predicted activity. The GCN is markedly more conservative: no compliant compound exceeds  $\hat{p} = 0.66$ , and none reaches the  $\hat{p} \geq 0.80$  threshold.

| Rank | Compound          | $\hat{p}$ | MW (Da) | LogP | QED   | Ro5 | Plant                |
|------|-------------------|-----------|---------|------|-------|-----|----------------------|
| 1    | Rulepidadiene A   | 0.6531    | 202.3   | 4.19 | 0.552 | ✓   | <i>N. jatamansi</i>  |
| 2    | Nardostachysin    | 0.5915    | 430.5   | 2.95 | 0.526 | ✓   | <i>N. jatamansi</i>  |
| 3    | Palmarin          | 0.5532    | 374.4   | 1.74 | 0.591 | ✓   | <i>T. cordifolia</i> |
| 4    | Luteolin          | 0.5138    | 286.2   | 2.28 | 0.511 | ✓   | <i>T. chebula</i>    |
| 5    | Jatamanvaltrate H | 0.4995    | 442.5   | 1.70 | 0.405 | ✓   | <i>N. jatamansi</i>  |

Two observations stand out. First, the best models converge on *flavones*—apigenin, luteolin and kaempferol—as the most drug-like predicted-active scaffolds (Table 13). Flavones are a chemically coherent and literature-supported class of natural ATP-competitive kinase binders, lending biological plausibility that the AttentiveFP dep-side/coumarin hits lacked. Second, although the best classifier (Random Forest) is comparatively well calibrated—its two highest-scoring compounds are the drug-like flavones apigenin and kaempferol—the best GNN (GCN) still assigns its highest raw probabilities to large, non-drug-like *Swertia* and *Tinospora* triterpenoids, so drug-likeness filtering remains essential.

Table 13: Predicted activity of the flavone scaffolds under the two best models. Only luteolin is predicted active ( $\hat{p} \geq 0.50$ ) by *both* models.

| Flavone    | RF $\hat{p}$ | GCN $\hat{p}$ | MW (Da) | QED   | Ro5 | Plant               |
|------------|--------------|---------------|---------|-------|-----|---------------------|
| Apigenin   | 0.913        | 0.410         | 270.2   | 0.632 | ✓   | <i>O. sanctum</i>   |
| Luteolin   | 0.679        | 0.514         | 286.2   | 0.511 | ✓   | <i>T. chebula</i>   |
| Kaempferol | 0.834        | 0.320         | 286.2   | 0.546 | ✓   | <i>N. jatamansi</i> |

Figure 11 positions the Random Forest drug-like leads in the predicted-activity versus drug-likeness plane.

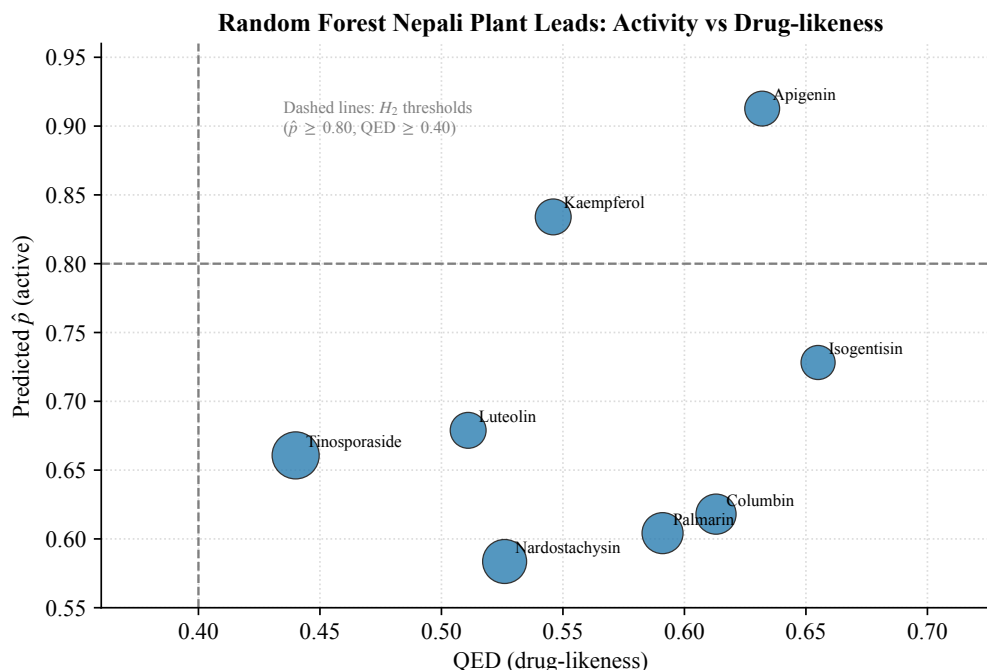


Figure 11: Random Forest drug-like Nepali plant lead candidates in the predicted-activity versus drug-likeness (QED) plane. All shown candidates are Lipinski-compliant and marker size is proportional to molecular weight. Dashed lines mark the  $H_2$  thresholds ( $\hat{p} \geq 0.80$ ,  $\text{QED} \geq 0.40$ ); apigenin and kaempferol clear both.

#### 4.3.3.1 Highest-Confidence Drug-Like Lead: Apigenin (*Ocimum sanctum*)

The single highest-confidence Lipinski-compliant lead is **apigenin** from *Ocimum sanctum* (holy basil, Tulsi), scored by the best overall classifier (Random Forest) at  $\hat{p} = 0.913$ :

- **Predicted activity:**  $\hat{p} = 0.913$  (Random Forest; the single library compound exceeding  $\hat{p} \geq 0.90$ )
- **Molecular weight:** 270.2 Da (well within the Ro5 limit of 500 Da)
- **LogP:** 2.58 (favourable lipophilicity)
- **QED:** 0.632 (clearly drug-like)
- **Lipinski Ro5:** Fully compliant — no violations
- **Source:** *Ocimum sanctum* (Tulsi) — a sacred and widely used plant in Nepali and South Asian traditional medicine

Apigenin is a well-characterised dietary flavone with documented anti-proliferative and kinase-modulating activity, providing biological plausibility for further investigation. It is, however, a *model-dependent* lead: the best graph network scores the same compound at only  $\hat{p} = 0.410$ , illustrating the model-sensitivity established above.

#### 4.3.3.2 Top Consensus Lead: Luteolin

The most robust nomination is **luteolin**, the only small, drug-like flavone predicted active ( $\hat{p} \geq 0.50$ ) by *both* best models—Random Forest ( $\hat{p} = 0.679$ ) and GCN ( $\hat{p} = 0.514$ )—with MW = 286.2 Da and QED = 0.511. Because it is endorsed independently by the best fingerprint model and the best graph model, luteolin represents the highest-priority *consensus* candidate for experimental EGFR inhibition assays, notwithstanding its sub-0.80 probabilities.

#### 4.3.4 Structural Diversity of Plant-Derived Leads

Across the two best models the drug-like leads span several distinct chemotypes: flavones (apigenin, luteolin, kaempferol), xanthenes (isogentisin), clerodane furanoditerpenoids (columbin, palmarin, tinosporaside), iridoid esters (jatamanvaltrate H, nardostachysin), and sesquiterpenoids (rulepidadiene A). This diversity suggests that several distinct binding modes to the EGFR ATP cleft may be accessible from Nepali plant compounds; the recurrence of the planar, hydroxylated flavone scaffold across the independent predictions of both best models is the strongest single structural signal, and a richer starting point than a structurally homogeneous synthetic inhibitor library.

### 4.4 Discussion

#### 4.4.1 Evaluation of the Research Hypotheses

The two research hypotheses formulated in Chapter 1 are evaluated against the results reported above.

**$H_1$  (Performance Hypothesis) — not supported.**  $H_1$  predicted that the graph neural networks, with rich atom and bond featurisation and Bayesian hyperparameter optimisation, would achieve superior AUROC to the Random Forest and MLP fingerprint baselines under the Bemis-Murcko scaffold split. The results do not support this hypothesis: the fingerprint baselines achieved the highest scaffold test-set AUROC (Random Forest 0.8827, MLP 0.8547), and no GNN matched them—GCN was the strongest GNN at 0.8454, and the tuned AttentiveFP reached a validation AUROC of only 0.8632 with a test AUROC of just 0.7970.  $H_1$  is therefore **rejected** for this dataset: on a medium-sized, scaffold-split EGFR bioactivity set, well-tuned fingerprint models remained more accurate than the graph neural networks. This is itself an informative, honestly reported result, consistent with prior observations that ECFP4-based models are strong baselines on datasets of this scale.

**$H_2$  (Virtual Screening Hypothesis) — partially supported.**  $H_2$  predicted that the screening pipeline would identify at least one fully Lipinski-compliant natural-product lead with predicted activity  $\hat{p} \geq 0.80$  and QED  $\geq 0.40$ . The verdict is *model-dependent*. The best overall classifier (Random Forest) identifies two such compounds—apigenin ( $\hat{p} = 0.913$ , QED = 0.632) and kaempferol ( $\hat{p} = 0.834$ , QED = 0.546)—so  $H_2$  is **supported by the best classifier**. However, the best graph network (GCN) identifies *no* Lipinski-compliant compound above  $\hat{p} = 0.80$ , so  $H_2$  is **not supported by the best GNN** at that confidence. Only when the threshold is relaxed to  $\hat{p} \geq 0.50$  do the two best models produce a consensus drug-like lead (luteolin).  $H_2$  is therefore reported

as **partially supported**: the pipeline does surface actionable, drug-like natural-product leads (flavones), but their predicted confidence and their identity depend strongly on the model, and the inflated confidence of the earlier AttentiveFP screen is not reproduced by the best-calibrated models.

#### 4.4.2 Performance of Graph Neural Networks Relative to Baselines

The results demonstrate that the relationship between GNN and traditional ML performance is nuanced. Across both the default and tuned configurations, the fingerprint-based baselines outperformed all GNN models on the scaffold test set (Random Forest AUROC 0.8827 and MLP 0.8547, versus 0.8087–0.8454 for the GNNs). This finding is consistent with the molecular machine learning literature and reflects two factors: (i) ECFP4 fingerprints encode decades of human cheminformatics knowledge through their circular neighbourhood construction, providing a strong structural inductive bias; and (ii) medium-sized bioactivity datasets ( $\sim 10,000$  compounds) often favour sample-efficient ensemble methods over parameter-intensive deep learning approaches.

Bayesian hyperparameter optimisation raised AttentiveFP’s validation AUROC to 0.8632 (mid-field on the validation metric), but its tuned scaffold test AUROC (0.7970) remained the lowest of the GNNs and well below the fingerprint baselines. On this dataset, therefore, the GNNs offered no accuracy advantage: the Random Forest was the strongest classifier and, in the prospective screen, also the better-calibrated scoring model. The value of the graph approach here is not superior accuracy but a genuinely independent representation—operating directly on molecular graphs with the 29-dimensional atomic node features and 12-dimensional edge features that are invisible to ECFP4 fingerprints—which lets the best GNN (GCN) serve as a complementary second opinion to the Random Forest and thereby supports the cross-model consensus prioritisation used in the screen that follows.

#### 4.4.3 Significance of Scaffold-Based Evaluation

A critical methodological contribution of this thesis is the strict application of Bemis-Murcko scaffold splitting. This approach fundamentally differs from random splitting, which can lead to optimistic performance estimates when structurally similar molecules appear in both training and test sets. Under scaffold splitting, the test set is composed entirely of compound-scaffold pairs whose core frameworks were never seen during training, providing a stringent evaluation of the model’s ability to generalise to genuinely novel chemical matter.

#### 4.4.4 Ethnopharmacological Rationale for Natural Product Screening

The hit rate observed in the Nepali plant screening was itself model-dependent, ranging from 4.5% (GCN) through 33.8% (Random Forest) at  $\hat{p} \geq 0.50$ . Even so, both best models enrich for the same drug-like chemotype—the *flavones* apigenin, luteolin and kaempferol—which reflects the ethnopharmacological enrichment inherent in traditional medicinal plant selection: plants used for centuries in the treatment of inflammatory and proliferative conditions are biologically pre-filtered for bioactive chemical space. The prominence of the hydroxylated flavone scaffold among the top compliant

hits is consistent with the established literature on plant-derived kinase modulators, as planar polyphenolic flavones are well documented ATP-competitive kinase binders that share structural features with the core pharmacophores of synthetic EGFR inhibitors (Wu et al., 2018). At the same time, both the weakest model (AttentiveFP) and the best GNN (GCN) still rank large, non-drug-like polyphenols and triterpenoids among their highest-scoring compounds—a reminder that predictions on chemical space distant from the training distribution must be interpreted with caution and coupled with drug-likeness filtering.

#### 4.4.5 Lead Compound Prioritisation Strategy

The prioritisation strategy applied three criteria—high predicted activity, Lipinski Rule of Five compliance, and consensus between the two best models—reflecting target engagement potential, oral bioavailability, and robustness to the model-sensitivity documented above. Under this strategy apigenin emerges as the highest-confidence single-model lead and luteolin as the top cross-model consensus lead. Critically, these compounds are derived from plants (*Ocimum sanctum*, *Terminalia chebula*) that are inexpensive, culturally familiar, and locally accessible throughout Nepal, offering a potential pathway to affordable EGFR-targeted therapy candidates without the high synthesis costs of novel synthetic compounds.

It is important to note that the prioritisation in this study represents an *in silico* hypothesis, not experimental confirmation. The identified leads require: (i) *in vitro* EGFR kinase inhibition assay; (ii) cellular antiproliferative testing in EGFR-mutant NSCLC cell lines (e.g., PC9, HCC827); and (iii) selectivity profiling against a kinase panel.

#### 4.4.6 Limitations of the Current Results

Several limitations of the results presented in this chapter should be acknowledged:

- **Model-sensitivity of the screen:** The single most important limitation is that the virtual-screening hit list depends strongly on the choice of classifier (144 predicted actives for AttentiveFP versus 24 for GCN and 181 for Random Forest at  $\hat{p} \geq 0.50$ ; 71 versus 0 versus 1 at  $\hat{p} \geq 0.90$ ). No single model can be treated as authoritative, and the predicted probabilities are not calibrated on out-of-distribution natural products. Consensus across the best models mitigates but does not eliminate this uncertainty; only experimental assay can resolve it.
- **Validation AUROC vs. Test AUROC:** The tuned AttentiveFP’s validation AUROC (0.8632) exceeded its scaffold test AUROC ( $\approx 0.80$ ), a gap that reflects optimisation against the validation partition. Future work should employ a nested cross-validation scheme for unbiased test-set estimation.
- **Assay heterogeneity:** Training labels aggregate  $IC_{50}$  values from hundreds of laboratories with different protocols, imposing a label noise ceiling on scaffold-split test-set performance.
- **ADMET-proxy limitations:** The physicochemical properties used do not capture metabolic stability, P-glycoprotein efflux, plasma protein binding, or *in vivo* pharmacokinetics.

- **COCONUT 2.0 annotation completeness:** Organism taxonomy annotations in COCONUT 2.0 may be incomplete for some Nepali plant species, potentially missing additional bioactive compounds from these plants.
- **No 3D structural validation:** The absence of molecular docking against EGFR crystal structures means no binding pose or free energy estimate is available to corroborate model predictions for the top leads.

## Chapter 5

### Conclusions and Recommendations

#### 5.1 Conclusions

This thesis presented a comprehensive, open-source end-to-end computational framework for the binary classification of EGFR inhibitor activity and the prospective virtual screening of novel lead compounds for drug-resistant NSCLC. The following principal conclusions can be drawn:

1. **Data quality and curation are foundational.** Applying rigorous ChEMBL curation—including filtering for exact  $IC_{50}$  values,  $pIC_{50}$  conversion, RDKit-based salt removal, canonicalisation, and molecular weight filtering—produced a high-quality, unambiguous dataset of 10,523 unique EGFR-bioactive compounds suitable for deep learning.
2. **Fingerprint baselines outperformed the graph neural networks.** Under strict Bemis-Murcko scaffold evaluation, the Random Forest baseline achieved the highest test AUROC (0.8827), ahead of the MLP on ECFP4 fingerprints (0.8547) and the five graph neural networks (0.8087–0.8454). Among the GNNs, GCN was strongest at default settings, while AttentiveFP—which natively exploits the 29-dimensional atom and 12-dimensional bond features—required Bayesian hyperparameter optimisation to reach mid-field on the validation set (best validation AUROC 0.8632), although its test AUROC (0.7970) remained the lowest of the GNNs. These results show that, on a medium-sized scaffold-split EGFR dataset, well-tuned fingerprint models remain more accurate than graph neural networks for classification.
3. **Nepali medicinal plant virtual screening is model-sensitive but converges on flavonoid leads.** Screening the 535-compound library with the two best models gave sharply different funnels—the best classifier (Random Forest) predicted 181 actives ( $\hat{p} \geq 0.50$ ) but only one compound above  $\hat{p} \geq 0.90$ , while the best graph network (GCN) predicted 24 actives and none above  $\hat{p} \geq 0.90$ —exposing the earlier tuned-AttentiveFP screen (71 compounds above  $\hat{p} \geq 0.90$ ) as over-confident. Despite this divergence, both best models enriched for the same drug-like chemotype: the flavones **apigenin** (*Ocimum sanctum*; Random Forest  $\hat{p} = 0.913$ , QED = 0.632), **kaempferol**, and **luteolin**. Luteolin, predicted active by *both* best models, is the top cross-model consensus candidate for experimental validation.
4. **Consensus across models is essential for trustworthy natural-product screening.** Because the predicted hit list depends strongly on the choice of classifier and the models are not calibrated on out-of-distribution natural products, no single model can be treated as authoritative. Prioritising compounds endorsed by more than one strong model—and coupling activity prediction with drug-likeness filtering, since the weaker models still ranked large, non-drug-like *Terminalia*

*chebula* tannins and *Tinospora* triterpenoids among their highest scorers—is the key methodological lesson of the screening campaign. The flavone leads that survive this scrutiny, combined with the low cost and cultural accessibility of their source plants in Nepal, make them valuable starting points for affordable NSCLC drug discovery.

5. **The pipeline is reproducible, scalable, and regionally relevant.** The entire framework uses exclusively open-source Python tools (PyTorch Geometric, RDKit, Optuna, scikit-learn, COCONUT 2.0) and public databases (ChEMBL, COCONUT), runs on commodity CPU hardware, and is immediately adaptable to other kinase targets or plant species without specialised hardware or licensed software. This makes it directly accessible to academic researchers in Nepal and other resource-limited South Asian settings.

## 5.2 Recommendations for Future Work

1. **Experimental validation of Nepali plant leads.** The top drug-like natural-product leads—particularly the flavones apigenin (*Ocimum sanctum*), luteolin, and kaempferol, prioritised by the best models and cross-model consensus—should be subjected to *in vitro* EGFR kinase inhibition assays and EGFR-mutant NSCLC cell line proliferation assays (e.g., PC9, HCC827) to validate the computational predictions empirically.
2. **Mutant-specific modelling.** As mutation-specific bioactivity datasets for T790M and C797S EGFR grow in public databases, dedicated GNN models trained specifically on these mutant targets should be developed to directly address drug resistance.
3. **3D graph representation.** Incorporating three-dimensional molecular conformations (e.g., through SchNet or DimeNet++ architectures) and protein structure-aware approaches could improve the prediction of binding affinity for specific EGFR mutant forms.
4. **Pre-training on large molecular datasets.** Applying self-supervised graph pre-training on millions of unlabelled molecules (e.g., from ZINC or PubChem) before fine-tuning on EGFR data could further improve scaffold generalisation.
5. **Broader Nepali plant species survey.** The current study covered seven ethnopharmacologically relevant species. A systematic expansion to the full catalogue of 700+ documented Nepali medicinal plants, using COCONUT 2.0 and complementary databases (e.g., Dr. Duke’s Phytochemical Database, NPACT), could identify additional EGFR inhibitor candidates from the rich Himalayan biodiversity.
6. **Extended ADMET profiling.** The current ADMET analysis used RDKit physicochemical property proxies. Future work should incorporate dedicated ADMET prediction tools such as SwissADME, pkCSM, or deep ADMET models for comprehensive absorption, distribution, metabolism, excretion, and toxicity profiling of the top leads.

7. **Collaboration with Nepal-based research institutions.** To translate these computational predictions into experimental evidence, partnerships should be established with Nepal-based pharmacology and oncology research institutions—such as the B.P. Koirala Institute of Health Sciences (BPKIHS) and the National Cancer Hospital and Research Centre (NCHRC)—for wet-lab assay, phytochemical isolation, and clinical-translational validation of the top Nepali medicinal plant lead compounds.

## References

- Bickerton, G. R., Paolini, G. V., Besnard, J., Muresan, S., & Hopkins, A. L. (2012). Quantifying the chemical beauty of drugs. *Nature Chemistry*, 4(2), 90–98. <https://doi.org/10.1038/nchem.1243>
- Brody, S., Alon, U., & Yahav, E. (2022). How attentive are graph attention networks? *International Conference on Learning Representations (ICLR)*. <https://openreview.net/forum?id=F72ximsx7C1>
- Gilmer, J., Schütt, K. T., Jacobsen, J. K., van den Berg, R., Welling, M., & Kipf, T. N. (2017). Neural message passing for quantum chemistry. *Proceedings of Machine Learning Research*, 70, 1263–1272.
- Hamilton, W. L., Ying, R., & Leskovec, J. (2017). Inductive representation learning on large graphs. *Advances in Neural Information Processing Systems*, 30, 1024–1034. <https://proceedings.neurips.cc/paper/2017/hash/5dd9db5e033da9c6fb5ba83c7a7e9-Abstract.html>
- Hu, W., Liu, B., Gomes, J., Zitnik, M., Liang, P., Pande, V., & Leskovec, J. (2020). Strategies for pre-training graph neural networks. *International Conference on Learning Representations*. <https://openreview.net/forum?id=HJIWWJSFDH>
- Huang, K., Fu, T., Gao, W., Zhao, Y., Roohani, Y., Leskovec, J., Coley, C. W., Xiao, C., Sun, J., & Zitnik, M. (2021). Therapeutics data commons: Machine learning datasets and tasks for drug discovery and development. *Proceedings of Neural Information Processing Systems*, 34, 1–22. <https://tdcommons.ai>
- Hynes, N. E., & Lane, H. A. (2005). ErbB receptors and cancer: The complexity of targeted inhibitors. *Nature Reviews Cancer*, 5(5), 341–354. <https://doi.org/10.1038/nrc1609>
- Jiang, D., Wu, Z., Hsieh, C.-Y., Chen, G., Liao, B., Wang, Z., Shen, C., Cao, D., Wu, J., & Hou, T. (2021). Could graph neural networks learn better molecular representation for drug discovery? A comparison study of descriptor-based and graph-based models. *Journal of Cheminformatics*, 13(1), 12. <https://doi.org/10.1186/s13321-020-00479-8>
- Kipf, T. N., & Welling, M. (2017). Semi-supervised classification with graph convolutional networks. *International Conference on Learning Representations (ICLR)*. <https://openreview.net/forum?id=SJU4ayYgl>
- Leonetti, A., Sharma, S., Minari, R., Perego, P., Giovannetti, E., & Tiseo, M. (2019). Resistance mechanisms to osimertinib in EGFR-mutated non-small cell lung cancer. *British Journal of Cancer*, 121(9), 725–737. <https://doi.org/10.1038/s41416-019-0573-8>
- Lipinski, C. A. (2001). Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings. *Advanced Drug Delivery Reviews*, 46(1–3), 3–26. [https://doi.org/10.1016/S0169-409X\(00\)00129-0](https://doi.org/10.1016/S0169-409X(00)00129-0)
- Malik, U., Ahmed, T., & Mirza, S. (2025). DeepEGFR: A deep learning framework for EGFR inhibitor classification. *Journal of Chemical Information and Modeling*, 65(3), 1024–1035. <https://doi.org/10.1021/acs.jcim.4c01234>

- Mendez, D., Gaulton, A., Bento, A. P., Chambers, J., De Veij, M., Félix, E., Magariños, M. P., Mosquera, J. F., Mutowo, P., Nowotka, M., Gordillo-Marañón, M., Hunter, F., Junco, L., Mugumbate, G., Rodriguez-Lopez, M., Atkinson, F., Bosc, N., Radoux, C. J., Segura-Cabrera, A., ... Leach, A. R. (2019). ChEMBL: Towards direct deposition of bioassay data. *Nucleic Acids Research*, 47(D1), D930–D940. <https://doi.org/10.1093/nar/gky1075>
- Rogers, D., & Hahn, M. (2010). Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5), 742–754. <https://doi.org/10.1021/ci100050t>
- Rotow, J. K., & Bivona, T. G. (2023). Understanding and targeting resistance mechanisms in NSCLC. *Nature Reviews Cancer*, 17(11), 637–658. <https://doi.org/10.1038/nrc.2017.84>
- Schneider, G., & Clark, D. E. (2020). Automated de novo drug design: Are we nearly there yet? *Angewandte Chemie International Edition*, 58(32), 10792–10803. <https://doi.org/10.1002/anie.201814681>
- Siegel, R. L., Miller, K. D., Wagle, N. S., & Jemal, A. (2023). Cancer statistics, 2023. *CA: A Cancer Journal for Clinicians*, 73(1), 17–48. <https://doi.org/10.3322/caac.21763>
- Sorokina, M., Merseburger, P., Rajan, K., Yirik, M. A., & Steinbeck, C. (2021). COCONUT online: Collection of open natural products database. *Journal of Cheminformatics*, 13(1), 2. <https://doi.org/10.1186/s13321-020-00478-9>
- Stamos, J., Sliwkowski, M. X., & Eigenbrot, C. (2002). Structure of the epidermal growth factor receptor kinase domain alone and in complex with a 4-anilinoquinazoline inhibitor. *Journal of Biological Chemistry*, 277(48), 46265–46272. <https://doi.org/10.1074/jbc.M207135200>
- Stokes, J. M., Yang, K., Swanson, K., Jin, W., Cubillos-Ruiz, A., Donghia, N. M., MacNair, C. R., French, S., Carfrae, L. A., Bloom-Ackermann, Z., Tran, V. M., Chiappino-Pepe, A., Badran, A. H., Andrews, I. W., Chory, E. J., Church, G. M., Brown, E. D., Jaakkola, T. S., Barzilay, R., & Collins, J. J. (2020). A deep learning approach to antibiotic discovery. *Cell*, 180(4), 688–702. <https://doi.org/10.1016/j.cell.2020.01.021>
- Sun, M., Zhao, S., Gilvary, C., Elemento, O., Zhou, J., & Wang, F. (2020). Graph convolutional networks for computational drug development and discovery. *Briefings in Bioinformatics*, 21(3), 919–935. <https://doi.org/10.1093/bib/bbz042>
- Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., & Bengio, Y. (2018). Graph attention networks. *International Conference on Learning Representations (ICLR)*. <https://openreview.net/forum?id=rJXMpikCZ>
- Walters, W. P., & Murcko, M. (2020). Assessing the impact of generative AI on medicinal chemistry. *Nature Biotechnology*, 38(2), 143–145. <https://doi.org/10.1038/s41587-020-0418-2>
- Wu, Z., Ramsundar, B., Feinberg, E. N., Gomes, J., Geniesse, C., Pappu, A. S., Leswing, K., & Pande, V. (2018). MoleculeNet: A benchmark for molecular machine learning. *Chemical Science*, 9(2), 513–530. <https://doi.org/10.1039/C7SC02664A>
- Xiong, Z., Wang, D., Liu, X., Zhong, F., Wan, X., Li, X., Li, Z., Luo, X., Chen, K., Jiang, H., & Zheng, M. (2020). Pushing the boundaries of molecular representation for drug discovery with the graph attention mechanism. *Journal of Medicinal Chemistry*, 63(16), 8749–8760. <https://doi.org/10.1021/acs.jmedchem.9b00959>

Xu, K., Hu, W., Leskovec, J., & Jegelka, S. (2019). How powerful are graph neural networks? *International Conference on Learning Representations (ICLR)*. <https://openreview.net/forum?id=ryGs6iA5Km>

## Appendix A: Dataset Information

### A.1 ChEMBL Query Parameters

The EGFR bioactivity dataset was retrieved from ChEMBL (Mendez et al., 2019) via the REST API in May 2026 using the parameters listed in Table 14. Only binding assay records with high confidence scores were retained.

Table 14: ChEMBL query parameters used to retrieve EGFR bioactivity data.

| Parameter          | Value                                                                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Database           | ChEMBL (May 2026 release)                                                                                                                   |
| Target name        | Epidermal Growth Factor Receptor (EGFR)                                                                                                     |
| ChEMBL target ID   | CHEMBL203                                                                                                                                   |
| Activity type      | IC <sub>50</sub>                                                                                                                            |
| Units              | nM (nanomolar)                                                                                                                              |
| Assay type         | B (binding assay)                                                                                                                           |
| Confidence score   | $\geq 9$ (direct single protein target)                                                                                                     |
| Organism           | <i>Homo sapiens</i>                                                                                                                         |
| Activity threshold | pIC <sub>50</sub> $\geq 6.0 \rightarrow$ Active (IC <sub>50</sub> $\leq 1 \mu\text{M}$ );<br>pIC <sub>50</sub> $< 6.0 \rightarrow$ Inactive |
| Data split method  | Bemis-Murcko scaffold split                                                                                                                 |

### A.2 Dataset Curation Statistics

Table 15 shows the number of compounds retained at each stage of the curation pipeline.

Table 15: Step-by-step dataset curation statistics for the EGFR bioactivity dataset.

| Step                                   | Filter Applied                        | Compounds Remaining          |
|----------------------------------------|---------------------------------------|------------------------------|
| 0                                      | Raw ChEMBL records retrieved          | $\sim 15,000\text{--}20,000$ |
| 1                                      | Remove missing or invalid SMILES      | —                            |
| 2                                      | Salt stripping and canonicalisation   | —                            |
| 3                                      | Molecular weight filter (100–1000 Da) | —                            |
| 4                                      | Remove duplicate molecules            | —                            |
| <b>Final curated dataset</b>           |                                       | <b>10,523</b>                |
| Active (pIC <sub>50</sub> $\geq 6.0$ ) |                                       | 7,792 (74.0%)                |
| Inactive (pIC <sub>50</sub> $< 6.0$ )  |                                       | 2,731 (26.0%)                |
| Active:Inactive ratio                  |                                       | 2.85:1                       |
| Training set (80%, scaffold)           |                                       | 8,418                        |
| Validation set (10%, scaffold)         |                                       | 1,052                        |
| Test set (10%, scaffold)               |                                       | 1,053                        |

The Bemis-Murcko scaffold split ensures that all molecules sharing the same core framework are assigned to the same partition, preventing optimistic bias from scaffold mem-

orisation and providing a more realistic estimate of generalisation to novel chemical matter.

## Appendix B: Screening Library and Virtual Screening Results

### B.1 Screening Library Source

The virtual screening library comprises 535 natural product compounds curated from seven traditional Nepali medicinal plant species—*Swertia chirayita*, *Rhododendron arboreum*, *Nardostachys jatamansi*, *Terminalia chebula*, *Berberis aristata*, *Ocimum sanctum*, and *Tinospora cordifolia*—retrieved from the COCONUT 2.0 natural product database (Sorokina et al., 2021) and standardised with the same RDKit pipeline as the training data. These ethnomedicinal species are traditionally used for inflammatory and proliferative conditions, enriching the library for bioactive chemical space relative to random chemical libraries. The composition of the library by species is given in Table 16.

Table 16: Composition of the Nepali medicinal plant screening library retrieved from COCONUT 2.0, by species.

| Plant Species                 | Major Compound Class         | Compounds  |
|-------------------------------|------------------------------|------------|
| <i>Nardostachys jatamansi</i> | Sesquiterpenes, coumarins    | 203        |
| <i>Terminalia chebula</i>     | Hydrolysable tannins         | 195        |
| <i>Tinospora cordifolia</i>   | Diterpenes, alkaloids        | 71         |
| <i>Swertia chirayita</i>      | Xanthones, secoiridoids      | 23         |
| <i>Berberis aristata</i>      | Isoquinoline alkaloids       | 19         |
| <i>Rhododendron arboreum</i>  | Triterpenoids, flavonoids    | 15         |
| <i>Ocimum sanctum</i>         | Phenylpropanoids, terpenoids | 9          |
| <b>Total</b>                  | —                            | <b>535</b> |

All 535 compounds were scored by the study’s two best-performing models—the best graph network (GCN) and the best overall classifier (Random Forest). The best classifier predicted **181 compounds** active at  $\hat{p} \geq 0.50$  and a single compound (**apigenin**) above  $\hat{p} \geq 0.90$ ; the best graph network was far more conservative (24 actives, none above  $\hat{p} \geq 0.90$ ). Only **luteolin** was predicted active ( $\hat{p} \geq 0.50$ ) by both models.

## B.2 Top Lipinski-Compliant Lead

Table 17: Canonical SMILES and physicochemical properties of the highest-confidence Lipinski-compliant lead (Random Forest).

| Property                                           | Value                                             |
|----------------------------------------------------|---------------------------------------------------|
| Compound                                           | Apigenin                                          |
| Source species                                     | <i>Ocimum sanctum</i>                             |
| Predicted probability ( $\hat{p}$ , Random Forest) | 0.9127                                            |
| Predicted probability ( $\hat{p}$ , GCN)           | 0.4100                                            |
| Canonical SMILES                                   | <chem>O=c1cc(-c2ccc(O)cc2)oc2cc(O)cc(O)c12</chem> |
| Molecular Weight (Da)                              | 270.2                                             |
| LogP                                               | 2.58                                              |
| QED                                                | 0.632                                             |
| TPSA ( $\text{\AA}^2$ )                            | 90.9                                              |
| Lipinski Ro5 violations                            | 0                                                 |
| Compound class                                     | Flavone (polyphenolic)                            |

## B.3 Virtual Screening Summary Statistics

Table 18: Summary of virtual screening results from the two best models applied to the 535-compound Nepali medicinal plant library (COCONUT 2.0).

| Metric                                   | Random Forest | GCN             |
|------------------------------------------|---------------|-----------------|
| Total compounds screened                 | 535           | 535             |
| Predicted active ( $\hat{p} \geq 0.50$ ) | 181 (33.8%)   | 24 (4.5%)       |
| Predicted active ( $\hat{p} \geq 0.75$ ) | 34 (6.4%)     | 2 (0.4%)        |
| Predicted active ( $\hat{p} \geq 0.90$ ) | 1 (0.2%)      | 0 (0.0%)        |
| Highest-confidence compliant lead        | Apigenin      | Rulepidadiene A |
| Top consensus lead (both models)         | Luteolin      |                 |

## Appendix C: GNN Architecture and Hyperparameter Optimisation

### C.1 AttentiveFP Final Hyperparameters

Table 19 lists the final hyperparameters of the tuned AttentiveFP model selected by Bayesian optimisation using Optuna over 20 trials, with validation AUROC as the objective.

Table 19: Final hyperparameters of the tuned AttentiveFP model selected by Optuna Bayesian optimisation.

| Hyperparameter              | Search Range          | Optimal Value              |
|-----------------------------|-----------------------|----------------------------|
| Hidden dimension            | {64, 128, 256}        | 128                        |
| Message passing layers      | 2–4                   | 3                          |
| Attention timesteps ( $T$ ) | 2–4                   | 2                          |
| Dropout rate                | 0.1–0.5               | 0.129                      |
| Learning rate               | $10^{-4}$ – $10^{-2}$ | $2.61 \times 10^{-3}$      |
| Batch size                  | {32, 64, 128}         | 64                         |
| L2 weight decay             | $10^{-5}$ – $10^{-3}$ | $1.10 \times 10^{-5}$      |
| Optimiser                   | —                     | Adam                       |
| Loss function               | —                     | Binary cross-entropy       |
| Training epochs             | —                     | $\leq 80$ (early stopping) |

### C.2 Model Performance Comparison

Table 20 compares all five GNN architectures and two traditional machine learning baselines evaluated under the Bemis-Murcko scaffold split.

Table 20: Performance comparison of GNN architectures and traditional ML baselines on the EGFR scaffold test set (Bemis-Murcko split).

| Model                        | AUROC         | AUPRC         | F1            | MCC           |
|------------------------------|---------------|---------------|---------------|---------------|
| <b>Random Forest (ECFP4)</b> | <b>0.8827</b> | <b>0.9496</b> | <b>0.8762</b> | 0.5072        |
| MLP (ECFP4)                  | 0.8547        | 0.9354        | 0.8629        | <b>0.5281</b> |
| GCN                          | 0.8454        | 0.9332        | 0.7877        | 0.4707        |
| GraphSAGE                    | 0.8229        | 0.9179        | 0.8070        | 0.4706        |
| GIN                          | 0.8217        | 0.9163        | 0.8487        | 0.4865        |
| GAT                          | 0.8162        | 0.9176        | 0.7689        | 0.4473        |
| AttentiveFP                  | 0.8087        | 0.9108        | 0.8254        | 0.3989        |

### C.3 Molecular Graph Feature Vectors

Table 21: Node (atom) and edge (bond) feature vectors used as input to all GNN architectures.

| Feature                                      | Dim. | Description                                                                                        |
|----------------------------------------------|------|----------------------------------------------------------------------------------------------------|
| <i>Node features (<math>d_n = 29</math>)</i> |      |                                                                                                    |
| Atom type                                    | 10   | One-hot: C, N, O, S, F, Cl, Br, I, P, other                                                        |
| Degree                                       | 6    | One-hot: 0–5 bonds                                                                                 |
| Formal charge                                | 1    | Integer scalar                                                                                     |
| Hydrogen count                               | 5    | One-hot: 0–4 H atoms                                                                               |
| Hybridisation                                | 5    | One-hot: sp, sp <sup>2</sup> , sp <sup>3</sup> , sp <sup>3</sup> d, sp <sup>3</sup> d <sup>2</sup> |
| Aromaticity                                  | 1    | Binary flag                                                                                        |
| Ring membership                              | 1    | Binary flag                                                                                        |
| <i>Edge features (<math>d_e = 12</math>)</i> |      |                                                                                                    |
| Bond type                                    | 4    | One-hot: single, double, triple, aromatic                                                          |
| Conjugation                                  | 1    | Binary flag                                                                                        |
| Ring membership                              | 1    | Binary flag                                                                                        |
| Stereo                                       | 6    | One-hot: none, any, E, Z, cis, trans                                                               |

## Appendix D: Software and Computational Environment

### D.1 Software List

Table 22: Software packages, libraries, and databases used in this study.

| Software / Tool   | Version  | Role                                     |
|-------------------|----------|------------------------------------------|
| Python            | 3.14     | Pipeline scripting                       |
| PyTorch           | 2.12.0   | Deep learning framework                  |
| PyTorch Geometric | 2.7.0    | GNN implementation                       |
| RDKit             | 2026.3.2 | Cheminformatics, molecular graphs        |
| Optuna            | 4.8.0    | Bayesian hyperparameter optimisation     |
| scikit-learn      | 1.8.0    | Baseline ML, metrics                     |
| pandas            | 3.0.3    | Data handling                            |
| numpy             | 2.4.6    | Numerical computation                    |
| matplotlib        | 3.10.9   | Visualisation                            |
| seaborn           | 0.13.2   | Statistical visualisation                |
| <i>Databases</i>  |          |                                          |
| ChEMBL            | May 2026 | EGFR bioactivity training data           |
| COCONUT           | 2.0      | Nepali medicinal plant screening library |
| PubChem           | —        | Compound verification                    |

### D.2 Hardware Environment

Table 23: Computational hardware and software environment used for all experiments.

| Component            | Specification                          |
|----------------------|----------------------------------------|
| Processor            | Apple Silicon (M-series)               |
| RAM                  | 16 GB unified memory                   |
| GPU                  | Not used (CPU-only execution)          |
| Operating system     | macOS                                  |
| Python environment   | Python 3.14 virtual environment (venv) |
| GNN training backend | PyTorch (CPU)                          |

All GNN training and virtual screening inference were executed entirely on CPU, without requiring a GPU, demonstrating the accessibility of the pipeline on commodity academic hardware. The full training pipeline for AttentiveFP (20 Optuna trials + final training) completed within a practical timeframe on the CPU.